

Machine-Learning-Assisted Empirical Bayes Estimation for Regression on Grouped Network Data

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Abstract

We propose a two-way effects model for matched data represented by a bipartite network with latent group structure. A directed acyclic graph (DAG) encodes the assumed relationships among the observed outcomes, network links, latent effects, group memberships, and prior parameters, thereby clarifying the data-generating process and organizing the empirical Bayes (EB) estimation procedure. Unobserved heterogeneity is modeled using group-specific prior distributions, allowing individual effects to vary both across and within groups. Although the two-way effects can be estimated by least-squares estimator, LS may be imprecise when the underlying bipartite network is sparse and connected across latent groups. We therefore develop a machine-learning-assisted EB estimator that first learns the latent group structure from the network and then exploits this structure to shrink the estimated effects toward their corresponding group-specific distributions. We establish the theoretical properties of the proposed estimator, and Monte Carlo simulations show that it substantially reduces estimation error relative to OLS under the data-generating processes considered. Finally, in an empirical application to Amazon user–product ratings, we use the proposed method to estimate and characterize heterogeneity in the user- and product-specific components of ratings.

Keywords: Two-way effect model, empirical Bayes method, directed acyclic graph, structural causal model, Bayesian belief network, group detection method, bipartite graph, matched data, shrinkage estimation.

JEL Codes: C11, C13, C18, C23, C55

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1 Introduction

Two-way fixed-effects regression has long been a workhorse for estimating high-dimensional heterogeneity in models based on interactions or matched data, dating back to the seminal contribution of [Abowd et al. \(1999\)](#). Important applications include the estimation of worker and firm effects ([Card et al., 2013](#)), student and teacher effects ([Jackson et al., 2014](#)), firm and bank effects in lending relationships ([Amiti and Weinstein, 2018](#)), patient and healthcare-location effects ([Finkelstein et al., 2016](#)), and family and neighborhood effects ([Chetty and Hendren, 2018](#)), among many others. More recently, the literature has connected conventional two-way effects models to network analysis by representing matched observations as a bipartite network. Within this framework, [Jochmans and Weidner \(2019\)](#) establish a direct connection between the statistical properties of least-squares estimators of two-way effects and the connectivity structure of the network induced by the matched data. In particular, weak network connectivity can substantially reduce the precision with which individual effects are estimated, generalizing the well-known limited-mobility problem in worker–firm applications. Motivated by this limitation, [Cheng et al. \(2025\)](#) develop a shrinkage approach and propose an empirical-Bayes estimator that regularizes the high-dimensional effects using a prior distribution informed by assortative matching patterns in the observed network. By borrowing information across nodes in the bipartite network, their estimator can improve the estimation of individual effects when the network provides insufficient information for precise unrestricted least-squares estimation.

This paper studies latent group structure in two-way effect model defined on bipartite network induced by matched data. Learning latent group structure from panel data has become an important topic in econometrics. A large literature develops grouped panel models in which cross-sectional heterogeneity is represented by a finite number of latent groups, with each observational unit assigned to one group; see, among others, [Bonhomme and Manresa \(2015\)](#); [Su et al. \(2016\)](#); [Bonhomme et al. \(2019\)](#); [Chetverikov and Manresa \(2022\)](#); [Mugnier \(2025\)](#). We contribute to this literature by introducing latent group structure on both sides of a two-way effect model, where the observed matching relationships form a group bipartite network and units on each side may belong to distinct latent communities. This structure provides an additional source of information that is ignored by the unrestricted least-squares estimator: although individual effects are heterogeneous, units belonging to the same latent group may share a common underlying distribution. The potential efficiency loss from ignoring such information is partic-

ularly important when the induced bipartite network is sparse or weakly connected, so that individual two-way effects are estimated from relatively limited information. Motivated by this observation, we propose an ML-assisted empirical-Bayes (EB) estimation framework that first learns the latent group memberships from the topology of the bipartite network using machine-learning-based community detection and then exploits the estimated group structure to construct group-specific prior distributions for the two-way effects. The resulting EB estimator selectively pools information among observational units belonging to the same estimated group while preserving within-group heterogeneity through individual-specific posterior estimates. Monte Carlo simulations show that the proposed estimator can substantially outperform the conventional least-squares estimator when sparsity and latent group structure coexist in the bipartite network induced by matched data.

Our first contribution is to introduce a network two-way effects model with latent group structure. Matched data naturally induce a bipartite network in which the two sets of nodes correspond to the two types of observational units and links represent observed matches between them. For example, in a student–teacher setting, a link is present between student i and teacher j whenever student i takes a course taught by teacher j . Connecting all observed student–teacher pairs therefore generates a bipartite network in which links occur only across the two node types, rather than among students or among teachers. [Jochmans and Weidner \(2019\)](#) exploit this network representation of matched data to study two-way effects and show that estimation of the individual effects is intrinsically related to the connectivity structure of the induced bipartite network. We extend this framework by allowing the bipartite network to exhibit a latent group structure on both sides. Such a structure is economically natural in many matching environments. For instance, students majoring in mathematics are more likely to take courses taught by mathematics instructors, while still taking some courses offered by other departments such as economics, computer science, or statistics. This generates relatively dense interactions between particular groups of students and instructors together with less frequent cross-group links. Similar patterns arise in worker–firm data, where positive assortative matching implies that more productive workers tend to be employed by more productive firms ([Abowd et al., 1999](#); [Kline, 2024](#)). These examples suggest that observed matches may contain systematic group-level structure beyond individual heterogeneity and overall network sparsity. Our model explicitly incorporates such latent group structure into the bipartite-network representation of the two-way effects model, allowing matching patterns across groups to provide

additional information about the heterogeneous effects on both sides of the network.

Our second contribution is to introduce a group-specific prior structure for the two sets of heterogeneous effects. Rather than imposing a single homogeneous prior distribution on all individual effects, we allow the distributions of the two-way heterogeneity to vary systematically across latent groups. Specifically, users belonging to the same user group share a common prior distribution for their individual effects, while products belonging to the same product group are governed by a corresponding group-specific prior. The latent network structure therefore determines how information is pooled across observational units in the empirical-Bayes estimation, while still allowing substantial heterogeneity among individuals within the same group. This specification is motivated by the distributional evidence in [Bonhomme et al. \(2019\)](#), who show in the worker–firm setting that workers and firms can be organized into groups according to worker quality and firm productivity and that matching frequencies vary systematically across such groups. In particular, relatively high-type worker groups tend to interact more frequently with relatively high-type firm groups, whereas lower-type groups display analogous matching patterns. We provide an explicit network representation of this idea by modeling the observed matches as a bipartite network with latent group structure and linking the estimated network memberships to group-specific prior distributions for the two-way effects. In this way, the proposed framework connects systematic group-level matching patterns in the network to the distributional heterogeneity of the underlying individual effects.

In contrast to conventional grouped fixed-effects models, which impose a common effect on all observational units belonging to the same latent group, we allow individual effects to remain heterogeneous within groups. In the standard grouped specification, $\alpha_i = \sum_{g=1}^G \mathbb{I}\{g_i = g\} \alpha_g$, where g_i denotes the group membership of unit i , α_g is the group-specific fixed effect, and $\mathbb{I}\{\cdot\}$ denotes the indicator function ([Bonhomme and Manresa, 2015](#); [Munro and Ng, 2022](#); [Chetverikov and Manresa, 2022](#)). Thus, conditional on group membership, all units within the same group are restricted to have an identical effect. We instead introduce a group-specific prior under which the individual effect α_i is drawn from a distribution whose parameters depend on its latent group. Formally, letting λ_g denote the set of distributional parameters associated with group g , we assume $\alpha_i \mid g_i \sim F\left(\cdot; \sum_{g=1}^G \mathbb{I}\{g_i = g\} \lambda_g\right)$. For example, under a Gaussian specification, $\alpha_i \mid g_i \sim \mathcal{N}\left(\sum_{g=1}^G \mathbb{I}\{g_i = g\} \mu_g, \sum_{g=1}^G \mathbb{I}\{g_i = g\} \sigma_g^2\right)$, where μ_g and σ_g^2 denote the group-specific prior mean and variance, respectively. This formulation retains the latent group structure emphasized in grouped regression models while relaxing the restriction of within-group

homogeneity. Consequently, heterogeneity is allowed to arise both between groups, through differences in the prior means and variances, and within groups, through individual-specific realizations from the corresponding group distributions. This additional flexibility is particularly useful for empirical-Bayes estimate, since it permits information to be pooled selectively within groups without forcing individual effects in the same group to be identical.

Our third contribution is to introduce a directed acyclic graph (DAG) as an integral component of the estimation framework. DAGs have long been used in econometrics and statistics to clarify model structure, conditional dependence, and the assumptions underlying regression specifications (Imbens, 2020; Imai and Kim, 2019). We go beyond this conventional interpretative role by using the DAG to organize the construction of the proposed shrinkage estimator. The procedure proceeds in several steps. First, we construct a DAG that provides a graphical representation of the data-generating process and the dependence relationships among the observed outcomes, heterogeneous effects, latent group memberships, and network structure. The corresponding regression model can then be represented through a structural model consistent with this graphical structure. Second, the DAG implies a factorization of the joint distribution into a sequence of lower-dimensional conditional distributions according to the local dependence relationships encoded by the graph. This probabilistic representation provides a Bayesian-network formulation of the model and makes explicit the components of the joint distribution that are relevant for inference on the heterogeneous effects. Third, we treat the posterior mean of the heterogeneous parameters as a class of shrinkage estimators. By Bayes' rule, their posterior distribution is proportional to the product of the likelihood implied by the outcome equation and the prior distribution of the heterogeneous effects. Finally, machine-learning methods are employed to learn the latent network group structure and thereby reduce the otherwise high-dimensional prior specification to a collection of group-specific distributions. Conditional on the estimated group memberships, the resulting posterior combines the information contained in the regression likelihood with that contained in the group-specific priors, yielding a tractable empirical-Bayes shrinkage estimator. In this sense, the DAG serves not only as a graphical device for interpreting the model but also as an organizing principle that connects the structural specification, probabilistic factorization, machine-learning step, and empirical-Bayes estimation within a unified framework.

The fourth contribution of this paper is to develop a specific machine-learning-assisted empirical-Bayes (EB) estimator for the proposed network two-way effects model with latent group struc-

ture. Although the conventional least-squares estimator can be applied to two-way effects models defined on matched networks, its precision depends critically on the connectivity and group structure of the underlying bipartite network (Jochmans and Weidner, 2019). When mobility or network connectivity is limited, individual effects may be estimated from relatively little independent information, leading to substantial estimation uncertainty. To address this problem, Cheng et al. (2025) propose an empirical-Bayes estimator that exploits shrinkage to trade bias for variance reduction, with the prior informed by assortative matching patterns observed in the network. Motivated by their use of network matching patterns to improve estimation, we exploit the latent group structure in the network by allowing the bipartite network to exhibit distinct communities on both sides and by allowing the distributions of the heterogeneous effects to vary systematically across these groups. The presence of latent group structure does not invalidate the least-squares estimator; rather, it provides additional information about cross-sectional heterogeneity that unrestricted least squares does not exploit. Building on the DAG-assisted empirical-Bayes framework introduced above, we use machine-learning methods to recover latent memberships on both sides of the bipartite network and subsequently estimate group-specific prior distributions for the corresponding heterogeneous effects. The resulting estimator combines the regression likelihood with these learned group-specific priors, shrinking each individual effect toward the distribution associated with its estimated group rather than toward a common population-wide distribution. Such selective pooling is particularly useful when limited network connectivity and systematic group-level matching coexist, as it exploits similarities among observational units while preserving heterogeneity both across and within latent groups.

We first present the main methodological components of the paper, including the proposed model, estimation procedure, and theoretical properties of the estimator. We establish identification of the two-way effect under an appropriate location normalization and connectedness of the bipartite network. We then characterize the relationship between the feasible empirical-Bayes estimator and its oracle counterpart. In particular, under consistent recovery of the latent group memberships and consistent estimation of the group-specific prior parameters, the feasible empirical-Bayes estimator asymptotically approaches the oracle estimator that uses the true group structure and prior parameters. Relative to conventional two-way least squares, the proposed estimator introduces shrinkage bias in exchange for variance reduction and can attain a lower mean squared error whenever the resulting reduction in variance dominates the increase in squared bias.

Next, we complement the theoretical analysis with a Monte Carlo study designed to reproduce two key features of the proposed framework: heterogeneous effects governed by latent group-specific distributions and a matched-data network exhibiting corresponding community structure. Specifically, we first assign units on both sides of the network to latent groups and generate their heterogeneous effects from group-specific Gaussian distributions. Conditional on these latent group memberships, we then generate the matching network according to a bipartite degree-corrected stochastic block model, following the degree-correction principle of [Karrer and Newman \(2011\)](#), so that the same latent memberships underlying the distributions of the heterogeneous effects also determine systematic patterns of matching in the network. Finally, conditional on the generated network and heterogeneous effects, we simulate the outcome observations from the proposed two-way effect model. This design allows us to systematically vary both the sparsity of the network and the strength of its latent group structure. Across the simulation designs, the proposed machine-learning-assisted empirical-Bayes estimator substantially reduces estimation error relative to conventional least squares ([Jochmans and Weidner, 2019](#)), with particularly pronounced gains when network connectivity is limited and the latent group structure is informative.

In the empirical application, we apply the proposed machine-learning-assisted empirical-Bayes estimator to study user and product heterogeneity in Amazon product ratings. We use the *Industrial and Scientific* category of the Amazon Reviews’23 dataset. We represent the observed user–product interactions as a bipartite network, in which users and products constitute the two distinct sets of nodes and a link is formed between user i and product j whenever user i leaves a rating for product j . This setting is particularly well suited to our framework because substantial heterogeneity may arise on both sides of the network, while many users and products are connected through only a limited number of observed interactions, making unrestricted estimation of high-dimensional two-way effects potentially imprecise. Our empirical objective is therefore to exploit the structure of the user–product network to recover latent groups on both sides and to use the resulting group memberships to guide information pooling in the subsequent empirical-Bayes estimation. Using only the pattern of observed user–product links, rather than the numerical rating outcomes themselves, the degree-corrected bipartite clustering procedure ([Rohe et al., 2016](#); [Zhou and Amini, 2019](#)) identifies 10 user groups and 11 product groups. The estimated group-specific prior distributions differ substantially in both their locations and dispersions, revealing systematic heterogeneity in users’ rating propensities and product-specific

rating effects across latent network communities. Incorporating this learned group structure into the empirical-Bayes estimator enables selective shrinkage of individual effects toward their corresponding group-specific distributions, thereby pooling information among users and products with similar network characteristics while preserving substantial heterogeneity both within and across groups.

This paper builds on several strands of the literature. First, it contributes to the literature on two-way effect model for matched data, originating with [Abowd et al. \(1999\)](#) and including subsequent applications and developments such as [Card et al. \(2013\)](#) and [Card et al. \(2016\)](#). Second, it relates to the literature on econometric models with latent group structure, in which parameters of interest are represented through a finite number of groups; see, among others, [Su et al. \(2016\)](#), [Bonhomme et al. \(2022\)](#), [Guðmundsson and Brownlees \(2021\)](#), and [Zhang et al. \(2019\)](#). Third, the paper builds on the empirical-Bayes and shrinkage literature, which uses information pooling to improve the estimation of high-dimensional heterogeneous parameters; relevant contributions include [Robbins \(1992\)](#), [Efron \(2012\)](#), [Ignatiadis and Wager \(2019\)](#), and [Efron \(2019\)](#). Fourth, our methodological framework draws on the literature on directed acyclic graphs, structural causal models, and Bayesian networks, which provides graphical and probabilistic representations of dependence structures and joint distributions; see, for example, [Pearl et al. \(2003\)](#), [Pearl \(2022\)](#), and [Koller \(2009\)](#). Finally, the paper is related to the growing literature on econometric and statistical models for network data, including [Zhu et al. \(2017\)](#), [Auerbach \(2022\)](#), and [Chen et al. \(2021\)](#). Our contribution lies at the intersection of these literatures by combining two-way heterogeneity, latent group structure in bipartite networks, machine-learning-based group recovery, and group-specific empirical-Bayes shrinkage within a unified estimation framework.

The rest of the paper is organized as follows. Section 2 presents the main model. Section 3 is devoted to estimation method. Section 4 is about theory. Section 5 and 6 are left to simulated and empirical studies, respectively. Finally, Section 7 concludes. Additional materials and proofs are provided in the Appendix and supplementary materials.

2 Network Two-Way Effect Regression Model with Group Structure

2.1 Main Model

In this subsection, we introduce the model. Let $\mathcal{G}_n = (\mathcal{S}_n, \mathcal{T}_n, \mathcal{E}_n)$ denote an observed bipartite network, where $\mathcal{S}_n$ and $\mathcal{T}_n$ are the two disjoint sets of nodes and $\mathcal{E}_n$ denotes the set of observed links. Let

$$N = |\mathcal{S}_n|, \quad M = |\mathcal{T}_n|, \quad n = |\mathcal{E}_n|.$$

For every observed link $(i, j) \in \mathcal{E}_n$, consider the two-way effect model for matched data

$$y_{ij} = \alpha_i^s + \alpha_j^t + x'_{ij}\beta + \varepsilon_{ij},$$

where α_i^s denotes the heterogeneous effect associated with node $i \in \mathcal{S}_n$, α_j^t denotes the heterogeneous effect associated with node $j \in \mathcal{T}_n$, x_{ij} is a vector of observed covariates, and ε_{ij} is an idiosyncratic disturbance.

Stacking observations over all links gives

$$Y = B\alpha + X\beta + \varepsilon,$$

where

$$\alpha = \begin{pmatrix} \alpha^s \\ \alpha^t \end{pmatrix} = \left(\alpha_1^s, \dots, \alpha_N^s, \alpha_1^t, \dots, \alpha_M^t \right)',$$

and B is the $n \times (N + M)$ bipartite incidence matrix. Each row of B corresponds to one observed link and contains a one in the column associated with its $\mathcal{S}_n$ -side node and a one in the column associated with its $\mathcal{T}_n$ -side node. Thus, if observation e corresponds to the link (i, j) ,

$$(B\alpha)_e = \alpha_i^s + \alpha_j^t.$$

Following the network two-way effect representation, define

$$\tilde{Y} = Y - X\beta.$$

Conditional on β , the model becomes

$$\tilde{Y} = B\alpha + \varepsilon. \quad (1)$$

We assume that the incidence matrix B exhibits a latent group structure. Since B is the matrix representation of the underlying bipartite network, we use the terms B and network interchangeably throughout the paper. Figure 1 illustrates this structure. Panel (a) shows a bipartite network in which units interact more frequently with units from corresponding groups—for example, economics students primarily take courses taught by economics instructors, while maintaining fewer links with instructors from other departments. Panel (b) presents the corresponding adjacency matrix after reordering nodes by group membership, revealing a clear block structure associated with the latent groups on both sides of the network.

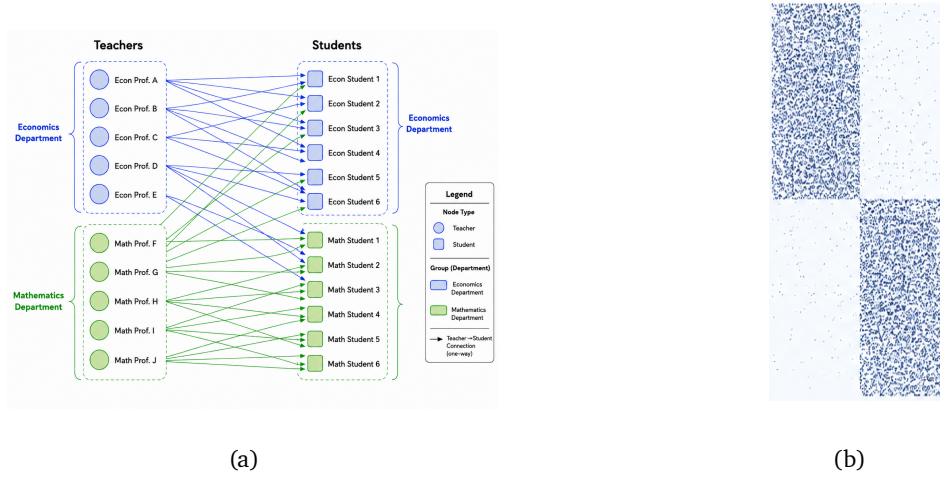


Figure 1: Panel (a) illustrates an example of a bipartite network with latent group structure on both sides of the network. Panel (b) displays the heatmap of the corresponding adjacency matrix after reordering the nodes according to their group memberships, revealing the resulting block structure.

Furthermore, we assume that the two sets of heterogeneous effects possess potentially different latent group structures. In particular, suppose that the $\mathcal{S}_n$ side contains K latent groups and the $\mathcal{T}_n$ side contains L latent groups. Let

$$g_i \in \{1, \dots, K\}, \quad h_j \in \{1, \dots, L\}$$

denote the respective memberships.

Rather than imposing a common fixed effect on all units belonging to the same group, we

assume that group membership determines the distribution of individual heterogeneity. Specifically,

$$\begin{aligned}\alpha_i^s \mid g_i = k &\sim \mathcal{N}\left(\mu_k^s, \tau_{k,s}^2\right), & k = 1, \dots, K, \\ \alpha_j^t \mid h_j = \ell &\sim \mathcal{N}\left(\mu_\ell^t, \tau_{\ell,t}^2\right), & \ell = 1, \dots, L.\end{aligned}$$

Hence units belonging to the same group share the same prior distribution but are not required to have identical heterogeneous effects. Equivalently, we may write

$$\begin{aligned}\alpha_i^s &= \mu_{g_i}^s + u_i^s, & \mathbb{E}[u_i^s \mid g_i] &= 0, \\ \alpha_j^t &= \mu_{h_j}^t + u_j^t, & \mathbb{E}[u_j^t \mid h_j] &= 0.\end{aligned}$$

Define means for both sides as

$$\mu = \left(\mu_{g_1}, \dots, \mu_{g_N}, \mu_{h_1}, \dots, \mu_{h_M}\right)',$$

and define variance as

$$\tau^2 = \left(\tau_{g_1}^2, \dots, \tau_{g_N}^2, \tau_{h_1}^2, \dots, \tau_{h_M}^2\right)'.$$

In addition to the main specification, we augment the proposed framework with a directed acyclic graph (DAG) that represents the underlying data-generating mechanism. Figure 2 illustrates the proposed generative structure and the associated dependence relationships among the model components. The DAG provides a graphical representation of how the latent group structure governs the distribution of individual heterogeneity and how this heterogeneity subsequently enters the outcome-generating process.

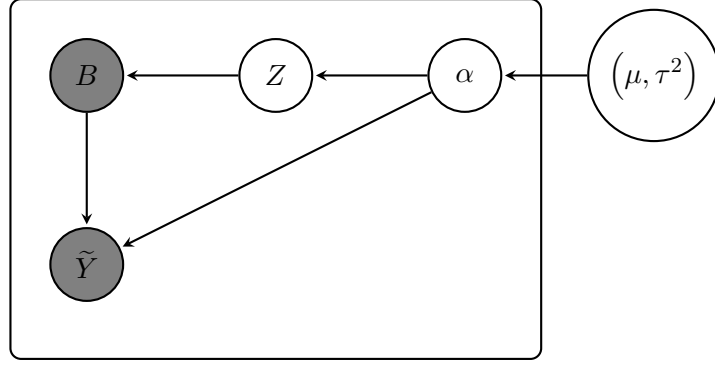


Figure 2: Graphical representation of the model in equation (1). Circles outside the plates denote hyperparameters, whereas circles inside the plates represent latent or observed variables. Shaded (gray) circles indicate observed quantities. In particular, the network B and the dependent variable Y are observed.

A directed acyclic graph can be associated with both a structural-equation representation and a probabilistic graphical representation. When the two representations share the same DAG, the structural equations describe how each variable is generated from its parent variables and an exogenous innovation, whereas the corresponding Bayesian network expresses the joint distribution through conditional distributions implied by the same parent–child relationships. In the present paper, we use this connection as an organizational device rather than relying on the intervention-based causal interpretation of a structural causal model. In particular, Figure 2 can first be represented by a system of structural equations describing the dependence relationships among the latent group memberships, heterogeneous effects, network structure, and observed outcome. The same DAG then implies a factorization of their joint distribution into a product of lower-dimensional conditional distributions. This factorization is useful for estimation because the posterior distribution of the heterogeneous effects can be obtained from the components of the joint distribution that depend on those effects. Consequently, the DAG plays two roles in our framework: it provides a graphical representation of the dependence structure of the model and, at the same time, organizes the probabilistic factorization that leads to the proposed empirical-Bayes estimator.

Formally, we introduce the following structural representation.

Definition 1. Consider the model in equation (1) together with the probabilistic graph in Figure 2.

The variables admit the structural representation

$$\begin{aligned}\alpha &= f_\alpha(\varepsilon_\alpha), \\ Z &= f_Z(\alpha, \varepsilon_Z), \\ B &= f_B(Z, \varepsilon_B), \\ \tilde{Y} &= f_{\tilde{Y}}(B, \alpha, \varepsilon_{\tilde{Y}}),\end{aligned}$$

where $\varepsilon_Z, \varepsilon_\alpha, \varepsilon_B, \varepsilon_{\tilde{Y}}$ denote exogenous innovations. The outcome equation is specified as

$$f_{\tilde{Y}}(B, \alpha, \varepsilon_{\tilde{Y}}) = B\alpha + \varepsilon_{\tilde{Y}},$$

and

$$f_\alpha = \varepsilon_\alpha,$$

where $\varepsilon_\alpha \sim \mathcal{N}(\mu, \text{Diag}(\tau^2))$, while $f_Z(\cdot)$ and $f_B(\cdot)$ are left unspecified.

The structural representation in Definition 1 induces the Bayesian-network factorization

$$\mathbb{P}(\tilde{Y}, B, \alpha, Z) = \mathbb{P}(\alpha) \mathbb{P}(Z \mid \alpha) \mathbb{P}(B \mid Z) \mathbb{P}(\tilde{Y} \mid B, \alpha). \quad (2)$$

Conditional on B and $\tilde{Y}$, we have

$$\mathbb{P}(\alpha \mid \tilde{Y}, B) \propto \sum_Z \mathbb{P}(\tilde{Y}, B, \alpha, Z). \quad (3)$$

The right-hand side of equation (3) can be replaced with the right-hand side of equation (2). In the feasible implementation, the latent membership Z is replaced by its machine-learning estimate $\hat{Z}$. Then, a sequence of conditional distribution can be further simplified into two terms where the first term is the likelihood implied by the regression equation, whereas the second term is the group-specific prior distribution of the heterogeneous effects. Furthermore, the unknown group-specific hyperparameters are estimated from the data. Then, the posterior mean

$$\hat{\alpha}^{EB} = \mathbb{E}[\alpha \mid \tilde{Y}, B, \hat{Z}]$$

then defines the proposed machine-learning-assisted empirical-Bayes shrinkage estimator. The details of the estimation procedure are deferred to the next section. In the remainder of this

section, we characterize the conditional distributions implied by the structural equations in Definition 1.

We explicitly specify the structural functions f_α and $f_{\tilde{Y}}$, since these two components are sufficient for constructing the empirical-Bayes estimator, while leaving f_Z and f_B unspecified to preserve flexibility in the latent group formation and network-generation mechanisms. The structural equations in Definition 1 induce corresponding conditional distributions. For example, under

$$\tilde{Y} = B\alpha + \varepsilon_{\tilde{Y}}, \quad \varepsilon_{\tilde{Y}} \sim \mathcal{N}(0, \sigma_\varepsilon^2 I_n),$$

the conditional distribution of the outcome is

$$\tilde{Y} \mid B, \alpha \sim \mathcal{N}(B\alpha, \sigma_\varepsilon^2 I_n).$$

Likewise, under the Gaussian group-specific prior,

$$\alpha \sim \mathcal{N}(\mu, D(\tau^2)),$$

where

$$D(\tau^2) = \text{diag}(\tau_1^2, \dots, \tau_{M+N}^2),$$

denotes the diagonal matrix of group-specific prior variances and μ denotes the vector of group-specific prior means and τ is the corresponding group-specific prior variances. Integrating out α yields

$$\tilde{Y} \mid B, Z \sim \mathcal{N}(B\mu, BD(\tau^2)B' + \sigma_\varepsilon^2 I_n).$$

Hence, the structural functions f_α and $f_{\tilde{Y}}$ directly determine the prior and likelihood components required for empirical-Bayes estimation. By contrast, because f_Z and f_B are left unspecified, the corresponding distributions governing latent group formation and network generation are also left unrestricted. In particular, under the DAG direction $\alpha \rightarrow Z$ and $Z \rightarrow B$, the distributions $\mathbb{P}(Z \mid \alpha)$ and $\mathbb{P}(B \mid Z)$ remain unspecified, whereas the group-specific prior $\mathbb{P}(\alpha)$ and the regression likelihood $\mathbb{P}(\tilde{Y} \mid B, \alpha)$ are specified explicitly.

Definition 2. Suppose that the model satisfies Definition 1. Then the structural equations imply

the following conditional distributions:

$$\begin{aligned}\alpha &\sim \mathcal{N}\left(\mu, D(\tau^2)\right), \\ Z \mid \alpha &\sim \mathbb{P}(Z \mid \alpha), \\ B \mid Z &\sim \mathbb{P}(B \mid Z), \\ \tilde{Y} \mid \alpha, B &\sim \mathcal{N}\left(B\alpha, \sigma_\varepsilon^2 I_n\right).\end{aligned}$$

Here,

$$D(\tau^2) = \text{diag}\left(\tau_1^2, \dots, \tau_p^2\right)$$

denotes the diagonal covariance matrix implied by the group-specific prior variances. Moreover,

$$\varepsilon_{\tilde{Y}} \sim \mathcal{N}\left(0, \sigma_\varepsilon^2 I_n\right).$$

The distributions $\mathbb{P}(Z \mid \alpha)$ and $\mathbb{P}(B \mid Z)$ are left unspecified according to Definition 1.

Definition 2 provides the probabilistic representation underlying the proposed empirical-Bayes estimation.

3 Machine-Learning-Assisted Empirical Bayes Estimation

In this subsection, we describe the estimation procedure for the proposed model. Conventional OLS-based estimators can be imprecise when the bipartite network B is sparse or connected in groups, while latent group structure may contain additional information about the heterogeneous effects that unrestricted least squares does not exploit. To address these limitations, we propose a novel empirical-Bayes estimator derived from the probabilistic graphical representation of the model and the factorization of its associated joint distribution. The model specification has been detailed in the preceding sections; here, we focus on the construction and implementation of the estimator.

The core component of the estimation procedure is the empirical-Bayes estimator, defined as the posterior conditional expectation of the unit-level heterogeneity. To derive this estimator, we begin with the joint distribution implied by the graphical model in Figure 2. Omitting the hyperparameters from the notation for brevity, the joint distribution in equation (2) can be decomposed further as follows:

$$\begin{aligned}\mathbb{P}(\alpha \mid \tilde{Y}, B) &\propto \sum_Z \mathbb{P}(\tilde{Y}, B, Z, \alpha) \\ &= \sum_Z \mathbb{P}(\tilde{Y} \mid \alpha, B) \mathbb{P}(B \mid Z) \mathbb{P}(Z \mid \alpha) \mathbb{P}(\alpha)\end{aligned}\tag{4}$$

$$\propto \sum_Z \mathbb{P}(\tilde{Y} \mid \alpha, B) \mathbb{P}(Z \mid B) \mathbb{P}(\alpha)\tag{5}$$

$$\begin{aligned}&\text{conditional on estimating } Z \text{ such that } \mathbb{P}(\hat{Z} \mid B) \rightarrow 1, \\ &\longrightarrow \mathbb{P}(\tilde{Y} \mid \alpha, B) \mathbb{P}(\alpha \mid \hat{Z}).\end{aligned}\tag{6}$$

where $\propto$ denotes proportionality. The equations above show how the posterior distribution of the heterogeneous effects is related to the likelihood and the group-specific prior. By Bayes' rule,

$$\mathbb{P}(\alpha \mid \tilde{Y}, B) \propto \mathbb{P}(\alpha, \tilde{Y}, B),$$

where the joint distribution on the right-hand side is obtained by integrating out the latent group memberships Z from the full joint distribution

$$\mathbb{P}(\tilde{Y}, B, Z, \alpha).$$

According to the graphical model in Figure 2, the full joint distribution factorizes into a sequence of conditional distributions, as shown in equation (2). Straightforward algebraic manipulation then yields equation (5).

Suppose further that the latent group structure can be consistently recovered from the observed network B , in the sense that

$$\mathbb{P}(\hat{Z} = Z \mid B) \rightarrow 1.$$

Under this condition, posterior uncertainty over the group memberships vanishes asymptotically, and equation (5) reduces to the feasible representation in equation (6), with the unknown Z replaced by $\hat{Z}$. The proposed empirical-Bayes estimator is then constructed as the posterior mean

$$\hat{\alpha}^{EB} = \mathbb{E}(\alpha \mid \tilde{Y}, B, \hat{Z}).$$

The transition from equation (4) to equation (5) shifts the focus from the generative distribution $\mathbb{P}(B \mid Z)$ to the inverse problem $\mathbb{P}(Z \mid B)$. Rather than modeling the network B as generated by latent group memberships Z , our objective is to infer the group structure given the observed network. A large body of literature has proposed algorithms for this task, many of which are consistent in the sense that the estimated group assignments $\hat{Z}$ satisfy $\mathbb{P}(\hat{Z} \mid B) \rightarrow 1$ as the network size grows.

Define group means as

$$\mu_g = \left(\mu_1^s, \dots, \mu_K^s, \mu_1^t, \dots, \mu_L^t \right)',$$

and define group variance as

$$\tau_g^2 = \left(\tau_1^2, \dots, \tau_K^2, \tau_1^2, \dots, \tau_L^2 \right)'.$$

Given the membership vectors $\hat{Z}$, the expression $\mu = \left(\mu_{g_1}^s, \dots, \mu_{g_N}^s, \mu_{h_1}^t, \dots, \mu_{h_M}^t \right)'$ can be rewritten compactly as $\hat{Z}\mu_g$. For unit i , $\hat{Z}_i$ is one-hot vector in which one entry is 1 indicating its group affiliation and the others are all 0. For example, $\hat{Z}_1 = [0, 1, 0]$ indicates that unit i belongs to group 2 with three groups in total. An analogous representation applies to the variance parameters σ_g . Therefore, while constructing EB estimator, the problem reduces from estimating μ and τ^2 to estimating μ_g and τ_g^2 conditiona on $\hat{Z}$.

Define the vector of group-specific prior means as

$$\mu_g = \left(\mu_1^s, \dots, \mu_K^s, \mu_1^t, \dots, \mu_L^t \right)',$$

and the vector of group-specific prior variances as

$$\tau_g^2 = \left((\tau_1^s)^2, \dots, (\tau_K^s)^2, (\tau_1^t)^2, \dots, (\tau_L^t)^2 \right)'.$$

Define the membership matrices

$$Z_s \in \{0, 1\}^{N \times K}, \quad Z_t \in \{0, 1\}^{M \times L},$$

where every row contains exactly one nonzero element indicating the group membership of unit

i. The joint membership matrix is

$$Z = \begin{pmatrix} Z_s & 0 \\ 0 & Z_t \end{pmatrix}.$$

Given the membership matrix Z , the vector of unit-specific prior means,

$$\mu = \left(\mu_{g_1}^s, \dots, \mu_{g_N}^s, \mu_{h_1}^t, \dots, \mu_{h_M}^t \right)',$$

can be written compactly as

$$\mu = Z\mu_g.$$

Note that $N + M \gg K + L$, where K is the number of groups among the units on the first-side of N and L is the number of groups among the units on the second-side of M .

An analogous representation applies to the unit-specific prior variances:

$$\tau^2 = Z\tau_g^2.$$

Hence, conditional on the group memberships Z , the empirical-Bayes estimation problem reduces from estimating the $(N + M)$ -dimensional vectors of unit-specific prior parameters μ and τ^2 to estimating the $(K + L)$ -dimensional vectors of group-specific hyperparameters μ_g and τ_g^2 .

3.1 Detecting Z Using Machine Learning

In this subsection, we discuss methods for estimating the latent membership matrix Z from the observed network B , and we denote the resulting estimator by $\hat{Z}$. Estimating Z is an important step in constructing Equation (6). Under the conditional-independence assumptions of the model, the plug-in posterior based on the estimated memberships satisfies

$$\mathbb{P}(\alpha \mid Y, B, \hat{Z}) \propto \mathbb{P}(Y \mid \alpha, B) \mathbb{P}(\alpha \mid \hat{Z}),$$

where $\mathbb{P}(Y \mid \alpha, B)$ is the likelihood and $\mathbb{P}(\alpha \mid \hat{Z})$ is the estimated group-specific prior.

Given B , our objective is therefore to apply a network-clustering method that consistently estimates the latent group assignments. Because group labels are identifiable only up to permu-

tation, an appropriate exact-recovery condition is

$$\mathbb{P}(\hat{Z} = ZP \text{ for some } P \in \mathcal{P}) \rightarrow 1,$$

where $\mathcal{P}$ denotes the set of admissible group-label permutation matrices.

Our framework can accommodate any group-detection algorithm satisfying the recovery condition and convergence rate required for the subsequent empirical Bayes analysis. One such approach is spectral clustering, which is among the most widely used and computationally efficient methods for detecting latent group structure in networks (Ng et al., 2001; Rohe et al., 2011, 2016). In the bipartite setting, the left and right singular vectors of a suitably normalized bi-adjacency matrix provide spectral embeddings for the two sets of nodes, which can then be clustered separately to estimate their respective memberships. Figure 3 illustrates examples of a bipartite stochastic block model under several parameter configurations.

For each parameter configuration illustrated in Figure 3, we generate 500 independent bipartite networks and apply the bi-spectral clustering algorithm to estimate the latent group memberships on both sides of the network. The table 1 summarizes the resulting recovery performance. The reported precision values are averaged across the 500 Monte Carlo replications after aligning the estimated group labels with the true labels. Overall, the results indicate that bi-spectral clustering effectively recovers the latent group structure under the data-generating processes considered.

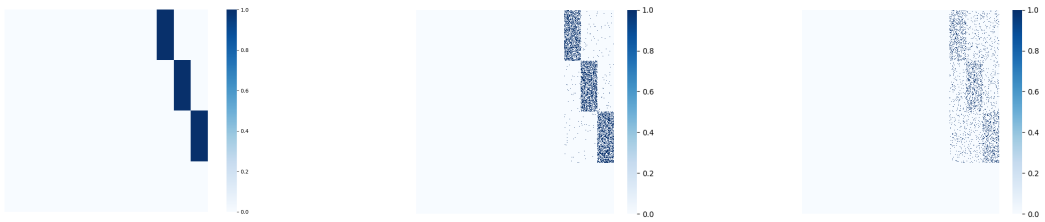


Figure 3: Three simulated bipartite networks with a common membership structure Z . The number of groups is fixed at three. Left: Bipartite network in which units within the same group are fully connected and there are no connections across groups. Middle: Bipartite network in which within-group connections are more likely than between-group connections. Right: Bipartite network generated under the same setup as the middle panel, but with weaker block structure, controlled by a lower contrast between within- and between-group connection probabilities.

	$p_{i,i}/p_{i,j} : 1.0/0.0$	$p_{i,i}/p_{i,j} : 0.5/0.01$	$p_{i,i}/p_{i,j} : 0.2/0.05$
precision	100%	100%	100%

Table 1: The table reports the classification precision of bi-spectral clustering applied to the simulated bipartite networks. Across all parameter configurations considered, spectral clustering achieves perfect classification accuracy when a block structure is present, with precision equal to 100%.

Table 1 reports a 100% exact-recovery rate for every network design considered: after optimal label alignment, the spectral clustering algorithm correctly classifies all nodes in each of the 500 Monte Carlo replications. Thus, within the simulated designs, the estimated memberships can be substituted for the true memberships without observed classification error, and the group-recovery condition underlying Equation (6) is satisfied empirically. These findings support the use of spectral clustering in the empirical application, particularly when its network structure resembles the well-separated group configurations examined in the simulations.

3.2 The Estimator

In this subsection, we build empirical Bayes estimator from equation (6).

The covariance matrix of α conditional on Z is

$$D(Z\tau_g^2) = \text{diag}(Z\tau_g^2).$$

Consequently,

$$\alpha \mid Z \sim \mathcal{N}(Z\mu_g, D(Z\tau_g^2)).$$

For the Gaussian benchmark, assume

$$\varepsilon \mid (B, Z, \alpha, X) \sim \mathcal{N}(0, \sigma_\varepsilon^2 I_n).$$

It follows that the marginal distribution of the residualized outcome is

$$\tilde{Y} \mid B, Z \sim \mathcal{N}(BZ\mu_g, BD(Z\tau_g^2)B' + \sigma_\varepsilon^2 I_n). \quad (7)$$

Equation (7) provides the basis for both identification of the group-specific hyperparameters and their maximum-likelihood estimation.

Consequently, conditional on the estimated group structure Z , the likelihood and prior jointly imply the following multivariate normal structure.

$$\begin{bmatrix} \tilde{Y} \\ \alpha \end{bmatrix} \Big| B, \hat{Z} \sim \mathcal{N} \left(\begin{bmatrix} B\hat{Z}\mu_g \\ \hat{Z}\mu_g \end{bmatrix}, \begin{bmatrix} BD(\hat{Z}\tau_g^2)B' + \sigma_\varepsilon^2 I_n & BD(\hat{Z}\tau_g^2) \\ D(\hat{Z}\tau_g^2)B' & D(\hat{Z}\tau_g^2) \end{bmatrix} \right). \quad (8)$$

Using standard results for conditional distributions of jointly normal random variables, the empirical Bayes estimator $\hat{\alpha} = \mathbb{E}(\alpha \mid B, \tilde{Y}, \hat{Z})$ admits the following closed-form expression:

$$\hat{Z}\mu_g + BD(\hat{Z}\tau_g^2) \left(BD(\hat{Z}\tau_g^2)B' + \sigma_\varepsilon^2 I_n \right)^{-1} (\tilde{Y} - B\hat{Z}\mu_g). \quad (9)$$

Then, the group-level parameters μ_g and τ_g can be estimated via marginal maximum likelihood. Following standard practice in the empirical Bayes literature, we obtain these estimates by maximizing the marginal likelihood function in equation (10).

$$\mathbb{P}(\tilde{Y} \mid B, \hat{Z}, \lambda) = \int_{\alpha} \mathbb{P}(\tilde{Y} \mid \alpha, B) \mathbb{P}(\alpha \mid \hat{Z}, \lambda) d\alpha, \quad (10)$$

which give us the estimated parameters $\hat{\lambda} = (\mu_g, \tau_g)$ as

$$\hat{\lambda}^{MLE} = \arg \max_{\lambda \in \mathcal{J}} \mathbb{P}(\tilde{Y} \mid B, \hat{Z}, \lambda). \quad (11)$$

Substituting the marginal maximum likelihood estimates $\hat{\lambda}^{MLE}$ into equation (9) yields the empirical Bayes estimator. We summarize the estimation procedure in the following algorithm.

Algorithm 1 ML-Assisted Empirical Bayes Estimation

Input: Y , X and incidence matrix B .

- 1: Using the algorithm, determine the number of groups $\hat{K}$ and $\hat{L}$ in B .
- 2: Using OLS, estimate $\hat{\beta}$ and calculate $\tilde{Y} = Y - X\hat{\beta}$.
- 3: Detecting group membership structure $\hat{Z}$ in B .
- 4: Construct EB estimator $\mathbb{E}(\alpha \mid B, \tilde{Y}, \hat{Z}, \lambda)$.
- 5: Estimate $\hat{\lambda}$ in $\mathbb{P}(\tilde{Y} \mid B, \hat{Z}, \lambda_g) = \int_{\alpha} \mathbb{P}(\tilde{Y} \mid \alpha, B) \mathbb{P}(\alpha \mid \hat{Z}, \lambda) d\alpha$.
- 6: Putting $\hat{\lambda}$ back to EB estimator.

Output: ML-Assisted Empirical Bayes estimator $\hat{\alpha}^{EB} = \mathbb{E}(\alpha \mid B, \tilde{Y}, \hat{Z}, \hat{\lambda})$.

Algorithm 1 summarizes the implementation of the proposed machine-learning-assisted empirical Bayes estimator. The observed data consist of an outcome vector Y , a matrix of covariates X . We additionally construct the edge-by-node design matrix B from the observed links, so that the two-way effects model can be written as

$$Y = B\alpha + X\beta + \varepsilon.$$

The first step is to estimate the numbers of first- and second-effect groups, denoted by K and L , respectively. These quantities determine the numbers of group-specific prior distributions: K distributions for the effects on one side of the network and L distributions for those on the other. Various procedures have been proposed for selecting the number of network communities (Le and Levina, 2015; Li et al., 2022; Karrer and Newman, 2011; Gao et al., 2018). Because community-number selection is not the primary focus of this paper, our framework permits any computationally feasible and theoretically justified selection procedure.

The second step is to estimate the slope coefficients β from a preliminary two-way effects regression that controls for the node effects represented by B . In particular, letting

$$M_B = I - B(B'B)^\dagger B',$$

where $(B'B)^\dagger$ denotes the Moore–Penrose generalized inverse, we compute

$$\hat{\beta} = (X'M_B X)^{-1} X'M_B Y$$

and construct the residualized outcome

$$\tilde{Y} = Y - X\hat{\beta}.$$

Here, β is a nuisance parameter estimated in the preliminary stage, while X is an observed covariate matrix; neither should be interpreted as a hyperparameter.

The third step is to estimate the latent membership matrix Z from the network B , yielding $\hat{Z}$. Any network-clustering procedure is admissible provided that it satisfies the recovery condition required by the subsequent empirical Bayes analysis. For example, if exact recovery is required,

the estimator should satisfy

$$\mathbb{P}(\hat{Z} = ZP \text{ for some admissible label permutation } P) \rightarrow 1.$$

Although our implementation uses spectral clustering, the empirical Bayes framework itself does not depend on a particular clustering algorithm.

Finally, given $\tilde{Y}$, B , and $\hat{Z}$, we construct the empirical Bayes estimator. Under the Gaussian prior specification, we first estimate the group-level hyperparameters

$$\lambda = (\mu_g, \tau_g^2)$$

by maximizing the marginal likelihood in Equation (10). Denoting the resulting marginal maximum-likelihood estimator by $\hat{\lambda}$, we substitute $\hat{\lambda}$ and $\hat{Z}$ into the posterior-mean expression in Equation (9). The resulting feasible posterior mean is the final empirical Bayes estimate of the heterogeneous node effects produced by Algorithm 1.

4 Theory

This subsection develops the identification conditions and theoretical properties of the proposed machine-learning-assisted empirical-Bayes (ML-EB) estimator. The analysis proceeds in four steps. First, we derive conditions for identifying the two-way heterogeneous effects and the associated group-specific prior parameters in the bipartite-network setting. Second, we provide conditions under which the latent group memberships on both sides of the network can be consistently recovered. Third, we establish consistency of the estimated prior hyperparameters and show that, under consistent group recovery and hyperparameter estimation, the feasible empirical-Bayes estimator asymptotically approaches its oracle counterpart. Finally, we characterize the mean squared error of the proposed estimator and use this characterization to clarify its bias–variance trade-off relative to the conventional least-squares estimator.

4.1 Identification

Identification of the proposed two-way effect model relies on two main conditions: an appropriate location normalization and connectedness of the observed bipartite network. Together, these conditions ensure that the two side unobserved heterogeneity are uniquely identified rel-

ative to the chosen normalization.

A fundamental identification issue in the two-way effect model is location indeterminacy. To see this, consider the residualized equation

$$\tilde{y}_{ij} = \alpha_i^s + \alpha_j^t + \varepsilon_{ij}.$$

For any scalar $c \in \mathbb{R}$, define

$$\alpha_i^{s,*} = \alpha_i^s + c, \quad \alpha_j^{t,*} = \alpha_j^t - c. \quad (12)$$

Then

$$\alpha_i^{s,*} + \alpha_j^{t,*} = \alpha_i^s + \alpha_j^t,$$

so the transformation in equation (12) leaves every conditional mean unchanged.

Define

$$v_0 = \begin{pmatrix} \mathbf{1}_N \\ -\mathbf{1}_M \end{pmatrix}.$$

Since each row of B contains one incidence from each side of the bipartite network, $Bv_0 = 0$. Hence v_0 belongs to the null space of B . A location normalization is therefore necessary.

Assumption 1 (Location normalization). *The group means on the $\mathcal{T}_n$ side satisfy*

$$\sum_{\ell=1}^L \pi_\ell^t \mu_\ell^t = 0,$$

where

$$\pi_\ell^t = \frac{M_\ell}{M}, \quad M_\ell = \sum_{j=1}^M \mathbf{1}\{h_j = \ell\}.$$

Moreover,

$$\mathbb{E}[\alpha_i^s - \mu_{g_i}^s \mid g_i] = 0 \quad \text{and} \quad \mathbb{E}[\alpha_j^t - \mu_{h_j}^t \mid h_j] = 0.$$

Assumption 1 fixes the relative location of the two sets of heterogeneous effects. The normalization could equivalently be imposed on the $\mathcal{S}_n$ side without affecting fitted values.

The second identification condition concerns the topology of the observed network. Let C_n denote the number of connected components of $\mathcal{G}_n$. For the bipartite incidence matrix used

above,

$$\text{rank}(B) = N + M - C_n.$$

Thus, every connected component generates one location indeterminacy.

Assumption 2 (Network connectivity). *The bipartite graph $\mathcal{G}_n$ is connected, so that*

$$C_n = 1$$

and consequently

$$\text{rank}(B) = N + M - 1.$$

Every node included in the analysis has strictly positive degree.

Under connectedness, the null space of B is one-dimensional and is spanned by v_0 . Hence the location normalization removes the only remaining indeterminacy.

Proposition 1 (Identification of the normalized two-way effects). *Suppose Assumptions 1–2 hold. Conditional on β , let α_1 and α_2 satisfy the same location normalization. If*

$$B\alpha_1 = B\alpha_2,$$

then

$$\alpha_1 = \alpha_2.$$

Thus the normalized two-way heterogeneous effects are point identified.

Importantly, introducing group-specific priors does not by itself remove this location indeterminacy when the prior means are estimated from the same data. In particular, consider the transformation

$$\begin{aligned} \alpha_i^{s,*} &= \alpha_i^s + c, & \text{and} & & \mu_k^{s,*} &= \mu_k^s + c, \\ \alpha_j^{t,*} &= \alpha_j^t - c, & \text{and} & & \mu_\ell^{t,*} &= \mu_\ell^t - c. \end{aligned}$$

The likelihood is unchanged, while

$$\alpha_i^{s,*} - \mu_{g_i}^{s,*} = \alpha_i^s - \mu_{g_i}^s \quad \text{and} \quad \alpha_j^{t,*} - \mu_{h_j}^{t,*} = \alpha_j^t - \mu_{h_j}^t.$$

Thus the group-prior density is also invariant. The empirical Bayes prior therefore provides regularization and partial pooling, but the location normalization remains necessary for identification.

The covariates must contain variation that is not perfectly explained by the two sets of node effects. Let P_B denote the projection matrix onto the identified column space of B , and define

$$M_B = I_n - P_B.$$

Assumption 3 (Covariate identification). *Let $p = \dim(\beta)$. The residualized design matrix has full column rank:*

$$\text{rank}(M_B X) = p.$$

Moreover, for some constant $c_X > 0$,

$$\lambda_{\min} \left(\frac{1}{n} X' M_B X \right) \geq c_X$$

with probability approaching one.

Assumption 3 excludes covariates that are perfectly collinear with the two-way heterogeneous effects. Under this condition, β is separately identified from α .

Remark 1 (Common intercept). *The baseline specification in (2.1) does not require a separate unrestricted common intercept because a constant vector lies in the span generated by the two sets of unrestricted effects. If the model is instead parameterized as*

$$y_{ij} = \beta_0 + \alpha_i^s + \alpha_j^t + x'_{ij}\beta + \varepsilon_{ij},$$

two location restrictions are required, for example

$$\sum_i w_i^s \alpha_i^s = 0, \quad \sum_j w_j^t \alpha_j^t = 0,$$

so that the common intercept β_0 is uniquely defined.

Conditional on the latent memberships,

$$\mathbb{E} [\tilde{y}_{ij} \mid g_i = k, h_j = \ell] = \mu_k^s + \mu_\ell^t.$$

Consequently, identification of the group means itself has a two-way structure.

Define a group-level bipartite graph whose first partition consists of the K groups on the $\mathcal{S}_n$ side and whose second partition consists of the L groups on the $\mathcal{T}_n$ side. Groups k and ℓ are connected if

$$\Pr((i, j) \in \mathcal{E}_n \mid g_i = k, h_j = \ell) > 0.$$

Assumption 4 (Group-level connectedness). *The induced (K, L) group-level bipartite graph is connected and every group contains at least one observed node.*

Under Assumptions 1 and 4, the group means

$$\mu_1^s, \dots, \mu_K^s, \mu_1^t, \dots, \mu_L^t$$

are identified from the first conditional moment. When the individual-level graph is connected and all groups are nonempty, its contraction into groups is also connected, so Assumption 4 is typically implied by Assumption 2.

Identification of the group-specific prior variances requires information from second moments. Let

$$\vartheta = \left(\tau_{1,s}^2, \dots, \tau_{K,s}^2, \tau_{1,t}^2, \dots, \tau_{L,t}^2, \sigma_\varepsilon^2 \right)'.$$

For $r = 1, \dots, K + L$, let D_r be a diagonal matrix that selects the nodes belonging to prior group r . Then

$$D(\tau^2) = \sum_{r=1}^{K+L} \tau_r^2 D_r,$$

and the marginal covariance matrix can be written as

$$\Omega(\vartheta) = \sum_{r=1}^{K+L} \tau_r^2 B D_r B' + \sigma_\varepsilon^2 I_n.$$

Assumption 5 (Variance-component identification). *The matrices*

$$\{B D_1 B', \dots, B D_{K+L} B', I_n\}$$

are linearly independent in the population limit. That is,

$$a_0 I_n + \sum_{r=1}^{K+L} a_r B D_r B' = 0$$

implies

$$a_0 = a_1 = \dots = a_{K+L} = 0.$$

Assumption 5 implies that distinct combinations of group-specific heterogeneity variances and idiosyncratic noise cannot generate the same covariance matrix. The economic intuition is particularly transparent from repeated links. For example, under conditional independence,

$$\text{Cov}(\tilde{y}_{ij}, \tilde{y}_{ij'} \mid g_i = k) = \tau_{k,s}^2, \quad j \neq j',$$

while

$$\text{Cov}(\tilde{y}_{ij}, \tilde{y}_{i'j} \mid h_j = \ell) = \tau_{\ell,t}^2, \quad i \neq i'.$$

Hence repeated observations sharing a node on either side of the network provide information that separates the corresponding heterogeneity variance from the idiosyncratic error variance. In contrast, a network in which all nodes have degree one generally provides insufficient covariance information for separating all variance components.

4.2 Recovery of the latent group structure

Given identification, consistency of the feasible empirical-Bayes estimator additionally requires consistent recovery of the latent group memberships and sufficient within-group information for consistent estimation of the group-specific prior means and variances, as well as the error variance.

The preceding identification arguments condition on Z . In practice, however, the latent memberships are unknown and must be recovered from the observed network. Let $\hat{Z}_s$ and $\hat{Z}_t$ denote the estimated memberships obtained from a consistent network clustering procedure, such as spectral clustering or singular-vector-based co-clustering.

Because the numerical labels of latent groups are arbitrary, consistency is defined up to label permutations. Let Π_s and Π_t denote permutation matrices of dimensions K and L , respectively.

Assumption 6 (Network community structure). *Conditional on the latent memberships, the bipartite adjacency matrix satisfies a degree-corrected stochastic block model*

$$A_{ij} \mid Z, \theta \sim \text{Bernoulli}(\theta_i^s \theta_j^t P_{g_i h_j}).$$

The numbers of groups K and L are fixed. Moreover, the limiting group proportions are bounded away from zero,

$$\frac{N_k}{N} \rightarrow \pi_k^s > 0, \quad \frac{M_\ell}{M} \rightarrow \pi_\ell^t > 0,$$

and distinct groups have distinct population connection profiles:

$$\begin{aligned} \min_{k \neq k'} \|P_{k,\cdot} - P_{k',\cdot}\| &\geq \delta_s > 0, \\ \min_{\ell \neq \ell'} \|P_{\cdot,\ell} - P_{\cdot,\ell'}\| &\geq \delta_t > 0. \end{aligned}$$

Finally, the expected node degrees increase sufficiently fast for the chosen spectral clustering procedure to consistently estimate the population singular subspaces.

For example, suppose

$$A_{ij} \mid Z, \theta \sim \text{Bernoulli} \left(\theta_i^s \theta_j^t P_{g_i h_j} \right).$$

The latent groups must generate sufficiently distinct connection profiles. A convenient separation condition is

$$\min_{k \neq k'} \|P_{k,\cdot} - P_{k',\cdot}\| \geq \delta_s > 0, \quad \text{and} \quad \min_{\ell \neq \ell'} \|P_{\cdot,\ell} - P_{\cdot,\ell'}\| \geq \delta_t > 0.$$

In addition, no relevant latent group is allowed to vanish asymptotically:

$$\frac{N_k}{N} \rightarrow \pi_k^s > 0, \quad k = 1, \dots, K, \quad \text{and} \quad \frac{M_\ell}{M} \rightarrow \pi_\ell^t > 0, \quad \ell = 1, \dots, L.$$

Together with an appropriate minimum-degree condition, these restrictions allow standard community-detection procedures to consistently recover the latent partitions.

Proposition 2 (Consistency of latent-group recovery). *Suppose Assumption 6 holds and the latent memberships are estimated using a consistent bipartite spectral clustering procedure. Then there exist permutation matrices Π_s and Π_t such that*

$$\begin{aligned} \frac{1}{N} \|\widehat{Z}_s - Z_s \Pi_s\| &\xrightarrow{p} 0, \\ \frac{1}{M} \|\widehat{Z}_t - Z_t \Pi_t\| &\xrightarrow{p} 0. \end{aligned}$$

Equivalently,

$$d_Z(\widehat{Z}, Z) \xrightarrow{p} 0,$$

up to a permutation of the group labels.

This result provides the formal justification for treating the recovered network communities as group-specific prior information in the second stage of the proposed ML-EB procedure.

4.3 Estimation and consistency of the prior hyperparameters

Let

$$\psi = (\mu', (\tau^2)', \sigma_\varepsilon^2)'$$

collect the unknown prior and disturbance parameters. Conditional on B and Z , define

$$\begin{aligned} m_\psi &= BZ\mu, \\ \Omega_\psi &= BD(\tau^2)B' + \sigma_\varepsilon^2 I_n. \end{aligned}$$

From (7), the marginal log likelihood, up to a constant, is

$$\begin{aligned} \ell_n(\psi; Z) &= -\frac{1}{2} \log |\Omega_\psi| \\ &\quad - \frac{1}{2} (\tilde{Y} - BZ\mu)' \Omega_\psi^{-1} (\tilde{Y} - BZ\mu). \end{aligned}$$

The feasible empirical Bayes hyperparameter estimator is

$$\widehat{\psi} = \arg \max_{\psi \in \Psi} \ell_n(\psi; \widehat{Z}).$$

Assumption 7 (Parameter space). *The true parameter ψ_0 is uniquely identified and belongs to the interior of a compact parameter space Ψ . In the regular interior case, there exist constants*

$$0 < \underline{\tau}^2 < \bar{\tau}^2 < \infty$$

and

$$0 < \underline{\sigma}^2 < \bar{\sigma}^2 < \infty$$

such that

$$\begin{aligned}\underline{\tau}^2 &\leq \tau_r^2 \leq \bar{\tau}^2, & r = 1, \dots, K + L, \\ \underline{\sigma}^2 &\leq \sigma_\varepsilon^2 \leq \bar{\sigma}^2.\end{aligned}$$

Allowing $\tau_r^2 = 0$ places the corresponding parameter on the boundary and requires a separate nonstandard asymptotic analysis. We therefore focus on the regular interior case.

Assumption 8 (Group growth). *For every $k = 1, \dots, K$ and $\ell = 1, \dots, L$,*

$$N_k \rightarrow \infty, \quad M_\ell \rightarrow \infty,$$

and the limiting group shares satisfy

$$\frac{N_k}{N} \rightarrow \pi_k^s > 0, \quad \frac{M_\ell}{M} \rightarrow \pi_\ell^t > 0.$$

Assumption 9 (Uniform convergence of the marginal likelihood). *For some deterministic sequence $a_n \rightarrow \infty$ and a deterministic criterion function $Q(\psi)$,*

$$\sup_{\psi \in \Psi} \left| \frac{1}{a_n} \ell_n(\psi; Z) - Q(\psi) \right| \xrightarrow{p} 0.$$

Moreover, $Q(\psi)$ is uniquely maximized at ψ .

Proposition 3 (Consistency of the estimated group priors). *Suppose Assumptions 1–9 hold. Suppose further that either exact group recovery holds or the remaining classification error has an asymptotically negligible effect on the normalized marginal likelihood, namely*

$$\sup_{\psi \in \Psi} \left| \frac{1}{a_n} \ell_n(\psi; \hat{Z}) - \frac{1}{a_n} \ell_n(\psi; Z) \right| = o_p(1).$$

Then, up to permutation of group labels,

$$\hat{\psi} \xrightarrow{p} \psi.$$

In particular,

$$\begin{aligned}\hat{\mu}_k^s &\xrightarrow{p} \mu_k^s, & \text{and} & & \hat{\tau}_{k,s}^2 &\xrightarrow{p} \tau_{k,s}^2, \\ \hat{\mu}_\ell^t &\xrightarrow{p} \mu_\ell^t, & \text{and} & & \hat{\tau}_{\ell,t}^2 &\xrightarrow{p} \tau_{\ell,t}^2,\end{aligned}$$

and

$$\hat{\sigma}_\varepsilon^2 \xrightarrow{p} \sigma_\varepsilon^2.$$

4.4 Oracle and feasible empirical Bayes estimators

Under the conditions above, the feasible estimator asymptotically approaches the corresponding oracle empirical-Bayes estimator that knows the true group assignments and hyperparameters.

Suppose first that the true group structure Z and true hyperparameters ψ are known. Define

$$m = Z\mu, \quad D = D(\tau^2).$$

The joint Gaussian structure implies that the oracle posterior mean of α is

$$\alpha^{EB,O} = m + DB' (BDB' + \sigma_\varepsilon^2 I_n)^{-1} (\tilde{Y} - Bm).$$

The feasible ML-EB estimator replaces Z_0 and ψ_0 by their estimated counterparts:

$$\hat{\alpha}^{EB} = \hat{m} + \hat{D}B' (B\hat{D}B' + \hat{\sigma}_\varepsilon^2 I_n)^{-1} (\tilde{Y} - B\hat{m}), \quad (13)$$

where

$$\hat{m} = \hat{Z}\hat{\mu}, \quad \hat{D} = D(\hat{\tau}^2).$$

Equation (13) makes the shrinkage mechanism explicit. The estimator starts from the group-specific prior mean $\hat{m}$ and updates it using the discrepancy

$$\tilde{Y} - B\hat{m}.$$

The magnitude of the update depends jointly on the within-group heterogeneity variances, the disturbance variance, and the topology of the bipartite network.

Theorem 1 (Oracle equivalence). *Let*

$$p_n = N + M$$

denote the total number of nodes in the bipartite network. Suppose that Propositions 2 and 3 hold.

In addition, suppose that:

- (i) the numbers of latent groups K and L are fixed;
- (ii) exact group recovery holds, up to label permutation,

$$\Pr\left(\widehat{Z} = Z\Pi\right) \rightarrow 1;$$

- (iii) the group-specific prior variances and the disturbance variance are uniformly bounded away from zero and infinity, i.e., for some constants

$$0 < \underline{\tau}^2 < \bar{\tau}^2 < \infty$$

and

$$0 < \underline{\sigma}^2 < \bar{\sigma}^2 < \infty,$$

$$\underline{\tau}^2 \leq \tau_r^2 \leq \bar{\tau}^2, \quad r = 1, \dots, K + L,$$

and

$$\underline{\sigma}^2 \leq \sigma_\varepsilon^2 \leq \bar{\sigma}^2;$$

- (iv) the average squared deviation of the heterogeneous effects from their group means is stochastically bounded,

$$\frac{1}{p_n} \|\alpha - m\|^2 = O_p(1), \quad m = Z\mu.$$

Then the feasible ML–EB estimator satisfies

$$\frac{1}{p_n} \left\| \widehat{\alpha}^{EB} - \alpha^{EB,O} \right\|^2 \xrightarrow{p} 0,$$

where $\alpha^{EB,O}$ denotes the oracle empirical Bayes estimator constructed using the true group memberships and the true hyperparameters.

Theorem 1 implies that estimation of the network groups and prior hyperparameters has an asymptotically negligible effect on the average squared distance between the feasible ML–EB estimator and its infeasible oracle counterpart. Hence the first-stage machine-learning procedure does not alter the limiting second-stage estimator when group recovery and hyperparameter estimation are sufficiently accurate.

4.5 Conditional mean squared error and the bias–variance trade-off

Relative to the conventional two-way least-squares estimator, empirical Bayes introduces shrinkage bias in exchange for a reduction in estimation variance. Accordingly, its conditional mean squared error is lower whenever the variance reduction dominates the increase in squared bias induced by shrinkage. In addition, under a correctly specified group-specific prior and quadratic loss, the posterior mean is the Bayes estimator and therefore minimizes integrated posterior risk. Consequently, its integrated Bayes risk is weakly lower than that of any alternative estimator, including unrestricted least squares, under the maintained prior specification.

The empirical Bayes estimator can also be characterized as a regularized two-way effects estimator. Let

$$m = Z\mu, \quad D = D(\tau^2),$$

and define

$$A = DB' (BDB' + \sigma_\varepsilon^2 I_n)^{-1}.$$

Then the oracle posterior mean can be written as

$$\hat{\alpha}^{EB} = m + A(\tilde{Y} - Bm).$$

Since

$$\tilde{Y} = B\alpha + \varepsilon,$$

we obtain

$$\begin{aligned} \hat{\alpha}^{EB} - \alpha &= m - \alpha + AB(\alpha - m) + A\varepsilon \\ &= -(I - AB)(\alpha - m) + A\varepsilon. \end{aligned}$$

Conditional on the realized heterogeneous effects α ,

$$\mathbb{E} [\hat{\alpha}^{EB} - \alpha \mid \alpha] = -(I - AB)(\alpha - m).$$

Thus empirical Bayes introduces shrinkage bias whenever $\alpha \neq m$. In contrast,

$$\text{Var} (\hat{\alpha}^{EB} \mid \alpha) = \sigma_\varepsilon^2 AA'.$$

The conditional mean squared error under quadratic loss is therefore

$$\begin{aligned} \text{MSE}_{EB}(\alpha) &= \|(I - AB)(\alpha - m)\|^2 \\ &\quad + \sigma_\varepsilon^2 \text{tr}(AA'). \end{aligned} \tag{14}$$

Equation (14) explicitly decomposes the estimation risk into squared shrinkage bias and sampling variance. The objective of the empirical Bayes procedure is not to eliminate bias, but rather to reduce variance sufficiently to obtain a lower overall mean squared error.

To illustrate how network topology affects this trade-off, consider the special case

$$D = \tau^2 I$$

on the identified subspace and define

$$\eta = \frac{\sigma_\varepsilon^2}{\tau^2}.$$

Let

$$Q_B = B'B.$$

Because the incidence matrix B contains two positive incidences per edge, Q_B is the signless Laplacian associated with the bipartite graph. For a bipartite graph it has the same eigenvalues as the usual graph Laplacian after an appropriate sign transformation on one partition.

Let

$$0 = q_1 < q_2 \leq \dots \leq q_{N+M}$$

denote the eigenvalues of Q_B under connectedness, with the zero eigenvalue corresponding to the location indeterminacy removed by the normalization. Let V denote the associated eigenvector matrix and define

$$b = V'(\alpha - m).$$

The conditional MSE of the empirical Bayes estimator becomes

$$\begin{aligned} \text{MSE}_{EB}(\alpha) &= \sum_{r=2}^{N+M} \frac{\eta^2 b_r^2}{(q_r + \eta)^2} \\ &\quad + \sigma_\varepsilon^2 \sum_{r=2}^{N+M} \frac{q_r}{(q_r + \eta)^2}. \end{aligned} \tag{15}$$

The first component of (15) is squared shrinkage bias, whereas the second is variance.

For the normalized least-squares estimator,

$$\text{MSE}_{OLS} = \sigma_\varepsilon^2 \sum_{r=2}^{N+M} \frac{1}{q_r}.$$

It follows immediately that empirical Bayes has lower conditional MSE than least squares if and only if

$$\sum_{r=2}^{N+M} \frac{\eta^2 b_r^2}{(q_r + \eta)^2} \leq \sigma_\varepsilon^2 \sum_{r=2}^{N+M} \left[\frac{1}{q_r} - \frac{q_r}{(q_r + \eta)^2} \right]. \quad (16)$$

Condition (16) provides a direct economic interpretation of the gain from group-based shrinkage. The left-hand side is the cost of shrinking the individual effects toward their group means, whereas the right-hand side is the resulting reduction in estimation variance. If units are relatively homogeneous within groups, then $\alpha - m$ is small and the shrinkage bias is limited. At the same time, if the network contains weakly identified directions, corresponding to small positive eigenvalues q_r , the variance of the conventional two-way least-squares estimator becomes large. The potential benefit of empirical Bayes is therefore greatest when the latent groups are informative and the bipartite network is sparse or weakly connected.

Proposition 4 (Conditional MSE dominance). *Suppose that $D = \tau^2 I$ on the normalized identified subspace and let*

$$\eta = \frac{\sigma_\varepsilon^2}{\tau^2}.$$

Let

$$Q_B = B' B$$

and denote its eigenvalues by

$$0 = q_1 < q_2 \leq \dots \leq q_{N+M},$$

where $q_1 = 0$ corresponds to the location indeterminacy removed by the normalization. Let

$$b = V'(\alpha - m),$$

where V is the orthogonal eigenvector matrix of Q_B and $m = Z\mu$.

Then the empirical Bayes estimator has weakly smaller conditional mean squared error than the

normalized least-squares estimator if and only if

$$\sum_{r=2}^{N+M} \frac{\eta^2 b_r^2}{(q_r + \eta)^2} \leq \sigma_\varepsilon^2 \sum_{r=2}^{N+M} \left[\frac{1}{q_r} - \frac{q_r}{(q_r + \eta)^2} \right].$$

Remark 2 (Variance reduction). *For every identified spectral direction $r \geq 2$ and every $\eta > 0$,*

$$\begin{aligned} \frac{1}{q_r} - \frac{q_r}{(q_r + \eta)^2} &= \frac{(q_r + \eta)^2 - q_r^2}{q_r(q_r + \eta)^2} \\ &= \frac{\eta(2q_r + \eta)}{q_r(q_r + \eta)^2} > 0. \end{aligned}$$

Thus empirical Bayes strictly reduces the variance contribution in each identified spectral direction. Conditional MSE dominance fails only when the squared shrinkage bias generated by $\alpha - m$ exceeds the resulting variance reduction.

This result also clarifies the distinction between identification and estimation precision. Network connectedness is sufficient to eliminate all relative location indeterminacies after normalization, but it does not guarantee precise estimation. A graph may be connected through only a small number of bridge links and consequently possess a very small q_2 . In this case the model remains point identified while the least-squares variance can nevertheless be large. The empirical Bayes prior regularizes precisely these weakly identified directions.

The preceding comparison conditions on the realized heterogeneous effects. A stronger optimality result follows when the risk is integrated with respect to the correctly specified prior distribution. Under squared error loss, the posterior mean minimizes posterior expected loss. Consequently, for any alternative estimator $\delta(\tilde{Y})$,

$$R_B(\hat{\alpha}^{EB}) \leq R_B(\delta),$$

where

$$R_B(\delta) = \mathbb{E} \left[\left\| \delta(\tilde{Y}) - \alpha \right\|^2 \right]$$

integrates over both the sampling distribution and the group prior. In particular,

$$R_B(\hat{\alpha}^{EB}) \leq R_B(\hat{\alpha}^{OLS}).$$

In the scalar-prior case, the Bayes risk has an especially transparent spectral representation.

Since

$$\eta = \frac{\sigma_\varepsilon^2}{\tau^2},$$

the posterior variance along spectral direction r is

$$\left(\frac{1}{\tau^2} + \frac{q_r}{\sigma_\varepsilon^2} \right)^{-1} = \frac{\sigma_\varepsilon^2}{q_r + \eta}.$$

Therefore,

$$R_B(\hat{\alpha}^{EB}) = \sigma_\varepsilon^2 \sum_{r=2}^{N+M} \frac{1}{q_r + \eta},$$

whereas

$$R_B(\hat{\alpha}^{OLS}) = \sigma_\varepsilon^2 \sum_{r=2}^{N+M} \frac{1}{q_r}.$$

Since $\eta > 0$ for a proper prior,

$$\frac{1}{q_r + \eta} < \frac{1}{q_r},$$

and hence

$$R_B(\hat{\alpha}^{EB}) < R_B(\hat{\alpha}^{OLS})$$

whenever the two estimators differ.

This result should be distinguished from Proposition 4. For a fixed realization of α , empirical Bayes need not dominate least squares because excessive shrinkage toward an inaccurate group mean can increase squared bias. However, when risk is integrated over the correctly specified group prior, the posterior mean is optimal under quadratic loss by construction.

The spectral representation also illustrates why empirical Bayes is particularly useful for sparse network data. Consider a direction for which

$$q_r \rightarrow 0.$$

The least-squares contribution to risk is

$$\frac{\sigma_\varepsilon^2}{q_r},$$

which diverges as $q_r \rightarrow 0$. In contrast, the corresponding empirical Bayes Bayes-risk contribution satisfies

$$\frac{\sigma_\varepsilon^2}{q_r + \eta} \rightarrow \frac{\sigma_\varepsilon^2}{\eta} = \tau^2.$$

The group prior therefore regularizes precisely those spectral directions that contain little information in the observed network.

This establishes a direct connection between network topology and the potential gain from empirical Bayes estimation. Connectedness determines whether the normalized effects are identified, while the magnitude of the positive eigenvalues of the network Laplacian determines the strength of identification and hence the sampling variability of conventional two-way fixed-effect estimators. The proposed group-specific empirical Bayes estimator exploits the estimated latent structure to stabilize these weakly identified directions through partial pooling.

5 Simulated Study

5.1 Simulation Design

In this section, we conduct a Monte Carlo simulation study to examine the finite-sample performance of the proposed machine-learning-assisted empirical Bayes (ML-EB) estimator. We consider a bipartite network with $K = L = 3$ latent groups on its two sides. First side contains 9,000 nodes, with 3,000 nodes assigned to each group and second side contains 3,000 nodes, with 1,000 nodes assigned to each group. Let $g_i \in \{1, 2, 3\}$ and $h_j \in \{1, 2, 3\}$ denote the group memberships of node i on side s and node j on side t , respectively. Their one-hot membership vectors are defined as

$$z_{s,i} = e_{g_i}, \quad z_{t,j} = e_{h_j},$$

where e_k denotes the k th canonical basis vector in $\mathbb{R}^3$.

We generate the node-specific effects from group-specific Gaussian distributions:

$$\alpha_{s,i} \mid g_i = k \sim \mathcal{N}(\mu_k^s, (\tau_k^s)^2), \quad \alpha_{t,j} \mid h_j = \ell \sim \mathcal{N}(\mu_\ell^t, (\tau_\ell^t)^2).$$

The group-specific means are set to

$$\mu^s = (1.0, 0.5, 0.0)', \quad \mu^t = (0.6, 0.3, 0.0)',$$

and all group-specific prior variances are set to

$$\tau_1^s = \tau_2^s = \tau_3^s = \tau_1^t = \tau_2^t = \tau_3^t = 0.1.$$

Equivalently, after stacking the effects and membership matrices from both sides, the prior distribution can be written as

$$\alpha \mid Z \sim \mathcal{N}\left(Z\mu_g, \text{diag}(Z\tau_g^2)\right).$$

The differences among the group-specific means generate separation across groups, while the positive prior variances preserve individual heterogeneity within each group.

Conditional on the group memberships, we generate the bipartite network using a degree-corrected stochastic block model. For each pair (i, j) , the link indicator satisfies

$$A_{ij} \mid Z, \Omega, \Theta \sim \text{Bernoulli}(p_{ij}), \quad p_{ij} = \theta_{s,i} \theta_{t,j} z_{s,i}' \Omega z_{t,j},$$

independently across node pairs. To isolate the role of group structure in the baseline design, we set

$$\theta_{s,i} = \theta_{t,j} = 1$$

for all i and j . The block-probability matrix is defined by

$$\Omega_{k\ell} = \begin{cases} 0.05, & k = \ell, \\ 0.001, & k \neq \ell. \end{cases}$$

Thus, links are substantially more likely to form between nodes belonging to groups with the same label than between nodes belonging to different groups.

For each observed link (i, j) , we independently generate one covariate and one idiosyncratic disturbance according to

$$x_{ij} \sim \mathcal{N}(0, 1), \quad \varepsilon_{ij} \sim \mathcal{N}(0, 1),$$

and set the slope coefficient to $\beta = 0.3$. The edge-level outcome is then generated as

$$y_{ij} = \alpha_{s,i} + \alpha_{t,j} + x_{ij}\beta + \varepsilon_{ij}, \quad (i, j) \in \mathcal{E}.$$

After stacking the outcomes associated with all observed links, the data-generating process can be expressed as

$$Y = B\alpha + X\beta + \varepsilon,$$

where B is the edge-by-node design matrix constructed from the simulated bipartite network.

5.2 Main Result

We conduct 500 independent Monte Carlo replications. In each replication, we first apply the bi-spectral clustering procedure described above to the simulated bi-adjacency matrix A . This produces the estimated membership vectors

$$\hat{z}_{s,i}, \quad i = 1, \dots, N,$$

for the nodes on side s , and

$$\hat{z}_{t,j}, \quad j = 1, \dots, M,$$

for the nodes on side t . We then stack these vectors to obtain the estimated membership matrix $\hat{Z}$. Next, we estimate β by OLS from the preliminary two-way fixed-effects regression and construct the residualized outcome

$$\tilde{Y} = Y - X\hat{\beta}.$$

Conditional on $\hat{Z}$ and the edge-by-node design matrix B , we estimate the group-specific prior means μ_g and variances τ_g^2 by maximizing the marginal likelihood in Equation (10). Finally, we substitute B , $\hat{Z}$, $\hat{\mu}_g$, and $\hat{\tau}_g^2$ into the posterior-mean expression in Equation (9) to obtain the feasible ML–EB estimator for that replication. The reported performance measures are calculated by averaging the corresponding results across the 500 replications.

Table 2: First part reports the setup and estimated means in group prior using marginal MLE. Second part reports the root mean square error of mean and median for α over 500 simulation replications, using $\frac{1}{N+M} \sum_{i=1}^{N+M} \sqrt{(\alpha_i - \hat{\alpha}_i)^2}$. μ_{h_3} is used as identification condition.

			$ML - EB_{MLE}$	
			Mean	Std.
μ_{g_1}	1.0	$\hat{\mu}_{g_1}$	0.99	$1.32 * 10^{-4}$
μ_{g_2}	0.5	$\hat{\mu}_{g_2}$	0.498	$4.814 * 10^{-5}$
μ_{g_3}	0.0	$\hat{\mu}_{g_3}$	0.002	$1.706 * 10^{-4}$
μ_{h_1}	0.6	$\hat{\mu}_{h_1}$	0.599	$2.21 * 10^{-5}$
μ_{h_2}	0.3	$\hat{\mu}_{h_2}$	0.301	$3.36 * 10^{-4}$
μ_{h_3}	0.0			

			$ML - EB_{MLE}$		OLS	
			Mean	Median	Mean	Median
$RMSE$			0.031	0.031	0.339	0.339

The first panel of Table 2 reports the results for estimating the group-level parameters using marginal maximum likelihood. $\mu_{h_3} = 0$ is used as identification condition. As shown in the first two columns, the true group means are set to $\mu_{g_1} = 1.0$, $\mu_{g_2} = 0.5$ and $\mu_{g_3} = 0$ on first side, and $\mu_{h_1} = 0.6$, $\mu_{h_2} = 0.3$ and $\mu_{h_3} = 0$ on second side. The fourth column reports the average estimated group means $\hat{\mu}$ across 500 simulation replications. The estimated values closely match the true parameter values used in the data-generating process. The fifth column reports the standard deviation of $\hat{\mu}$ across the 500 simulations, indicating that the estimators are both accurate and stable.

The second panel of Table 2 presents the performance of the proposed machine-learning-assisted empirical Bayes (ML-EB) estimator. For comparison, we also report results from the ordinary least squares (OLS) estimator. Estimation accuracy is evaluated using the root mean squared error (RMSE), $\frac{1}{N+M} \sum_{i=1}^{N+M} \sqrt{(\alpha_i - \hat{\alpha}_i)^2}$. As shown in the final row of Table 2, the ML-EB estimator achieves a substantially lower average RMSE than the OLS estimator. This result indicates

that group-specific shrinkage improves the recovery of the underlying heterogeneous effects under the data-generating processes considered. The second-panel results therefore demonstrate the effectiveness and finite-sample advantages of the proposed ML–EB estimator relative to unrestricted OLS.

6 Empirical Study

6.1 Amazon User Ratings

Online product ratings have become an important source of information for consumers, firms, and researchers studying digital markets. Existing research documents that Amazon ratings and reviews can affect product sales, consumers’ acquisition of information, and consumer welfare (Chevalier and Mayzlin, 2006; Reimers and Waldfogel, 2021). Other studies emphasize that observed ratings may also be shaped by reviewer identity, selective reviewing, and the sequential evolution of consumer opinions (Godes and Silva, 2012; Forman et al., 2008). Accordingly, an observed rating may reflect not only the underlying appeal of a product but also the systematic rating tendency of the user providing the evaluation: some users consistently assign relatively high ratings, whereas others evaluate products more critically. Nevertheless, much of the existing literature relies on product-level average ratings or focuses on particular characteristics of reviews, leaving the two-sided heterogeneity underlying individual user–product evaluations relatively underexplored. In this subsection, we represent Amazon ratings as observations generated on a bipartite network connecting users and products and estimate a model incorporating both user- and product-specific heterogeneity,

$$Rating_{(i,j),t} = Intercept + User_{i,t} + Product_{j,t} + Innovation_{(i,j),t}.$$

The model decomposes each observed rating into a common component, a persistent user-specific rating tendency, a product-specific appeal component, and an idiosyncratic residual, thereby providing a more granular characterization of rating variation on both sides of the online marketplace.

We further allow the user–product bipartite network to exhibit latent group structure. In particular, users with similar interests may rate overlapping sets of products, while products with related characteristics may attract similar populations of users. For example, consumers

interested in cooking may disproportionately rate kitchen products, students may concentrate on educational supplies and electronic devices, and sports enthusiasts may be more likely to rate sporting goods. These examples motivate, rather than impose, the presence of group structure; the latent groups are learned from the observed network using a data-driven procedure. Recovering this structure allows us to estimate group-specific prior distributions of user and product effects through empirical Bayes shrinkage. This approach can improve estimation precision in sparse networks, characterize heterogeneity both within and across latent market segments, and potentially provide information relevant to marketing strategies directed toward particular consumer groups.

Our empirical analysis uses the Industrial and Scientific category of the Amazon Reviews’23 dataset compiled by the McAuley Lab at the University of California, San Diego. Amazon Reviews’23 is a large-scale collection of product reviews and user–product interactions spanning numerous product categories and an extended period. The complete Industrial and Scientific category contains approximately 5.2 million ratings submitted by 3.4 million users for approximately 427,500 products. Rather than using these raw data, we employ the officially provided 5-core, de-duplicated (*Pure IDs*) version. The 5-core procedure iteratively removes users and products with fewer than five distinct interactions, thereby retaining a denser portion of the bipartite network in which every included node has repeated connections. De-duplication removes duplicate review records and retains a single observation for each included user–product pair. These preprocessing steps substantially mitigate weak connectivity problem and limit the influence of users and products that appear only infrequently. This restriction is particularly useful in our setting because estimation of user- and product-specific heterogeneity requires sufficient overlap between the two sides of the network, while the precision of node-specific estimates generally deteriorates for low-degree nodes. The resulting estimation sample contains 412,947 distinct user–product rating observations involving 50,985 unique users and 25,848 unique parent products. The reviews in the estimation sample were posted between July 2001 and August 2023.

Before estimating the heterogeneous user and product effects, we first recover the latent group structure of the observed user–product interaction network. Importantly, this first-stage classification uses only the existence of user–product interactions and does not use the numerical rating outcome. This separation prevents the dependent variable used in the subsequent two-way fixed-effect and empirical-Bayes analysis from mechanically determining the latent groups.

We represent the data as an undirected and unweighted bipartite network and let $(B \in 0, 1^{N \times M})$ denote its bi-adjacency matrix, where (N) and (M) are the numbers of users and products, respectively, and $B_{ij} = 1$ user i rated product j . Since the data are de-duplicated, each observed user–product pair contributes one link regardless of the numerical value of the rating. In our Amazon Industrial and Scientific sample, the resulting network contains $(N=50,985)$ users, $(M=25,848)$ products, and $(E=412,947)$ observed user–product links. Moreover, because the sample is constructed using the 5-core restriction, every retained user and product has repeated interactions and hence strictly positive degree, so the degree normalization introduced below is well defined.

Before estimating the heterogeneous user and product effects, we first estimate this latent group structure from the observed user–product interaction network. Importantly, the first-stage classification uses only the presence or absence of user–product links and does not use the corresponding numerical rating outcomes. This separation avoids outcome-driven group construction and ensures that the dependent variable in the subsequent two-way fixed-effects and empirical Bayes analysis does not mechanically determine the estimated group memberships. The estimated groups should consequently be interpreted as interaction-based segments: users are grouped according to the products they rate, and products are grouped according to the users from whom they receive ratings. Let’s define A as the adjacency matrix. Thus, $(A_{ij} = 1)$ indicates the presence of an observed interaction, whereas $(A_{ij} = 0)$ indicates that no rating by user (i) for product (j) is observed. Because the data are de-duplicated, each observed user–product pair contributes exactly one edge, irrespective of the numerical rating assigned. Moreover, under the 5-core restriction, every retained user and product is incident to at least five distinct links. Consequently, all nodes have strictly positive degree, and the degree-based normalization introduced below is well defined. We use the corrected BIC criterion to determine the number of groups [Hu et al. \(2020\)](#),

$$\ell(K, L) = \max_{z, \theta} \log f(A \mid \theta, z) - [\lambda(N \log K + M \log L) + KL \log NM] .$$

We then select

$$(\hat{K}, \hat{L}) = \arg \max_{K, L} \ell(K, L).$$

Applying the group-number selection procedure to the empirical network yields $\hat{K} = 10$ user groups and $\hat{L} = 11$ product groups. Given these selected numbers of groups, we construct spectral embeddings from the left and right singular vectors of the degree-normalized bi-adjacency

matrix and apply clustering separately to the user and product embeddings (Rohe et al., 2016). Figure 4 presents the resulting group structure. Panel (a) displays the user–product bipartite network. Panel (b) displays the bi-adjacency matrix after its rows and columns have been reordered according to the estimated user and product memberships, respectively. The reordered matrix exhibits pronounced blockwise variation in link density, indicating that particular user groups interact disproportionately with particular product groups. Each rectangular block represents the interactions between one user group and one product group.

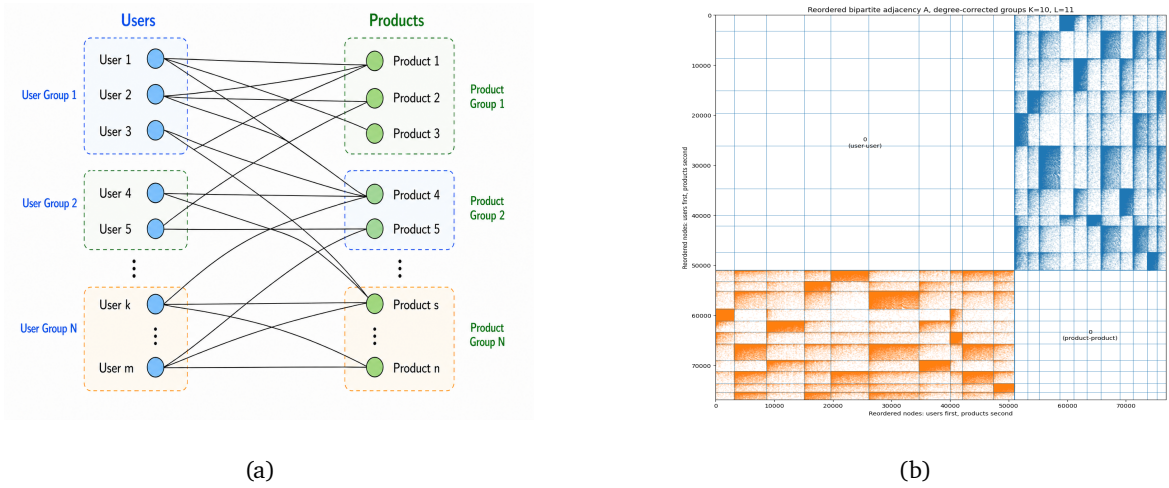


Figure 4: Panel (a) illustrates a user–product ratings bipartite network in which particular groups of users tend to interact more frequently with particular groups of products. Panel (b) displays the heatmap of the corresponding biadjacency matrix, with both rows and columns reordered according to the estimated cluster memberships obtained from the bipartite spectral clustering procedure.

Next, given estimated group memberships, we estimate unobserved heterogeneity. The empirical Bayes procedure exploits the estimated group structure of the user–product network by allowing the distributions of the user and product effects to vary across their respective latent groups. We adopt group-specific Gaussian priors for both sets of effects. Because users are partitioned into 10 groups and products into 11 groups, the specification contains 21 group-specific prior distributions: 10 for the user effects and 11 for the product effects. This structure accommodates heterogeneity both among individual users and products within the same group and across network-defined groups through differences in the locations and dispersions of their prior distributions. Under the normalization imposed on the user and product effects, the estimated common intercept is 4.425 stars. Table 3 reports the estimated group-specific prior means and standard deviations.

The estimates reveal substantial heterogeneity in rating behavior across user groups. In par-

Table 3: Estimated group-specific prior parameters.

User Groups				Product Groups			
Group	N_{nodes}	Prior Mean	Prior SD	Group	N_{nodes}	Prior Mean	Prior SD
1	3,152	0.1513	0.1644	1	2,252	-0.0128	0.2830
2	5,531	0.0437	0.4134	2	1,903	0.0095	0.2617
3	6,429	-0.1719	0.6965	3	3,576	0.0834	0.2391
4	4,515	0.0311	0.4574	4	2,387	0.1100	0.1220
5	6,525	0.0062	0.4998	5	2,241	-0.1717	0.3932
6	8,558	0.0439	0.4219	6	2,375	-0.0257	0.2367
7	5,295	-0.0701	0.5973	7	3,270	0.0876	0.2171
8	2,085	0.1130	0.3045	8	2,188	-0.1824	0.4027
9	5,314	0.0188	0.4568	9	2,473	0.0249	0.2694
10	3,581	-0.0020	0.4830	10	1,655	0.0313	0.2859
				11	1,528	0.0343	0.2531

ticular, the estimated prior means of the user effects range from -0.172 for User Group 3 to 0.151 for User Group 1. Because the dependent variable is measured in rating points, these estimates have a direct interpretation under the adopted normalization. Holding the product-specific effect constant, a user in Group 1 assigns ratings that are, on average, approximately 0.151 points above the common benchmark, whereas a user in Group 3 assigns ratings approximately 0.172 points below it. The difference between the two group-specific prior means is therefore 0.323 rating points. This result suggests that the groups inferred exclusively from the interaction network also capture substantively meaningful differences in rating behavior. One possible interpretation is that Group 1 contains relatively favorable or generous reviewers, whereas Group 3 contains comparatively demanding or critical reviewers. Because the model estimates product-specific effects simultaneously, these differences are not mechanically attributable to some user groups reviewing products with systematically higher or lower average ratings. Nevertheless, the estimates describe conditional rating tendencies and should not be interpreted as causal effects of group membership.

The estimated group-specific prior variances reveal an additional dimension of heterogeneity in user rating behavior. User Group 1 has an estimated prior variance of only 0.027 , corresponding to a prior standard deviation of approximately 0.164 rating points, whereas User Group 3 has an estimated prior variance of 0.485 , corresponding to a standard deviation of approximately 0.696 . Thus, Group 1 is characterized not only by a relatively high prior mean but also by comparatively homogeneous user effects. In contrast, the effects of users in Group 3 are much more widely dispersed around the group’s negative prior mean, indicating substantial within-group

variation in rating tendencies. This contrast illustrates an important advantage of the group-specific empirical Bayes approach. Imposing a common prior distribution on all users would obscure both differences in average rating tendencies across groups and differences in within-group dispersion. By estimating a separate prior distribution for each group, the procedure allows the shrinkage pattern to adapt to group-specific heterogeneity. Holding the information available for an individual effect fixed, a smaller prior variance produces stronger shrinkage toward the corresponding group mean, whereas a larger prior variance preserves more individual-level variation. In the joint two-way effects model, the actual amount of posterior shrinkage also depends on the number and pattern of observed ratings and on the dependence induced by the bipartite network design.

A similarly pronounced pattern emerges for the product effects. The estimated product-group prior mean is lowest for Product Group 8 at approximately -0.182 , followed closely by Product Group 5 at -0.172 , and is highest for Product Group 4 at approximately 0.110 . Under the adopted normalization and holding user-specific rating tendencies constant, products in Group 4 have a systematically higher average product-specific rating component than products in Groups 5 and 8. In particular, the difference between the prior means of Product Groups 4 and 8 is 0.292 rating points. This result suggests that the product groups inferred from the bipartite interaction network are informative about systematic differences in how products are evaluated. The estimated group structure may capture similarities in product characteristics, market segments, user-product matching patterns, or other latent attributes associated with product ratings. Importantly, the groups are estimated exclusively from the binary interaction structure rather than from the numerical ratings or observed product characteristics. The association between these network-defined groups and the estimated product effects therefore indicates that reviewing patterns contain useful information about product-side rating heterogeneity. Nevertheless, the product effects should be interpreted as conditional rating components rather than as causal measures of product quality.

The estimated product-group prior variances provide an additional dimension of product-side heterogeneity. Product Group 4 has an estimated prior variance of only 0.015 , corresponding to a prior standard deviation of approximately 0.122 rating points. This dispersion is substantially smaller than the estimated variances of 0.155 and 0.162 for Product Groups 5 and 8, which correspond to standard deviations of approximately 0.394 and 0.402 , respectively. The relatively small prior variance for Group 4 indicates that the product effects in this group are tightly con-

centrated around their group-specific mean, making that mean particularly informative about an individual product’s latent effect. Holding product-level information and network position fixed, the empirical Bayes estimator therefore shrinks the effects of products in Group 4 more strongly toward their group mean. By contrast, the larger prior variances for Groups 5 and 8 indicate substantial product-level heterogeneity even after conditioning on group membership, leading the estimator to preserve more of the individual variation in product effects. These findings are consistent with a flexible hierarchical representation in which network-defined groups provide useful information about product similarity without imposing identical effects on all products within the same group.

Empirical Bayes shrinkage is particularly important in the present application because node degrees vary considerably across users and products, even within the 5-core sample. An unrestricted two-way fixed-effects estimator treats every user and product effect as a separate parameter and imposes no distributional pooling across nodes. Consequently, effects associated with relatively few or weakly informative network connections may be estimated with considerable noise. The empirical Bayes estimator addresses this problem by combining the information contained in each node’s observed ratings with the estimated prior distribution of its assigned group. All else being equal, an imprecisely estimated individual effect is shrunk more strongly toward its group-specific prior mean, whereas a precisely estimated effect receives greater weight from its individual data and is shrunk less. The amount of shrinkage depends not only on the number of observed ratings but also on the configuration of network links and the estimated variance of the relevant group-specific prior. Crucially, shrinkage is not directed toward a single population-wide mean. A user assigned to User Group k is shrunk toward the estimated prior mean of that user group, whereas a product assigned to Product Group ℓ is shrunk toward the corresponding product-group mean. Thus, prior-induced information sharing occurs primarily among nodes with similar network interaction patterns rather than through indiscriminate pooling across all users or products. At the same time, the joint posterior continues to account for the dependence between user and product effects induced by the observed bipartite network.

Taken together, the results indicate that the network-based group structure contains substantively relevant information about heterogeneity on both sides of the Amazon user–product rating network. Although numerical ratings are not used to construct the groups, users assigned to different network-defined groups exhibit systematically different rating tendencies and degrees of within-group dispersion. Similarly, product groups differ in both their average product-

specific rating components and their within-group heterogeneity. Incorporating this structure into the empirical Bayes estimator permits selective partial pooling: information is shared primarily among users or products assigned to the same group, while distinct node-specific effects are retained. The proposed estimator therefore combines the flexibility of two-way individual effects with the potential precision gains from group-specific regularization, providing a natural framework for estimating high-dimensional, two-sided heterogeneity in sparse bipartite networks. These findings should nevertheless be interpreted as descriptive evidence of systematic heterogeneity rather than as causal effects of group membership. The estimated groups are not exogenously assigned treatments; instead, they arise from observed interaction patterns that may reflect self-selection, matching behavior, user preferences, product attributes, and other unobserved factors.

7 Conclusion

We propose a novel two-way effect model driven by bipartite network induced by matched data. The model is assisted using a directed acyclic graph that not only explicitly characterizes the latent data-generating mechanism but lead to a new framework of empirical Bayes estimate. Building on this DAG for proposed model, we thus develop a specific machine-learning-assisted empirical Bayes estimator, motivated by the poor performance of ordinary least squares in networks that are sparse and exhibit latent group structure. We establish the identification and theoretical results characterizing the properties of the proposed estimator. In addition, we introduce a novel group prior that allows for within-group heterogeneity while preserving latent group structure. An application of the proposed methodology to the Amazon User-Product ratings data reveal a group-structured user and product heterogeneity, which assist us to build more precise and distinct marketing strategies targeting at different groups of users.

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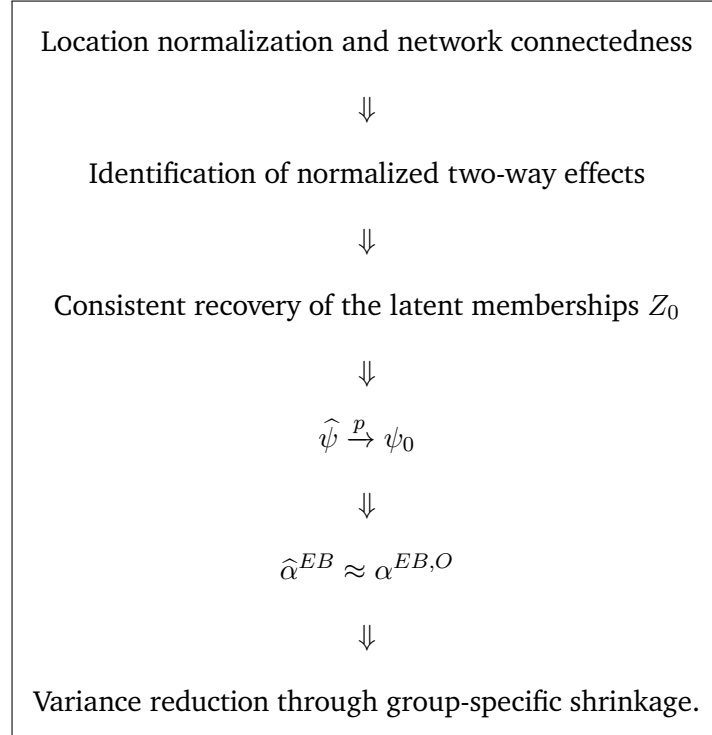
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Appendix A. Proof

Summary of the Theoretical Argument. The theoretical results above imply the following sequence:



Thus, the group prior does not replace the network conditions required for identification. Rather, connectedness and normalization identify the relative two-way effects, consistent network clustering recovers the relevant latent types, and empirical Bayes uses these types to pool information selectively across observationally similar nodes. The resulting estimator trades a controlled amount of shrinkage bias for a potentially substantial reduction in variance, with the largest gains arising in sparse or weakly connected parts of the bipartite network. $\square$

Proof of Proposition 1. Since

$$B(\alpha_1 - \alpha_2) = 0,$$

Assumption 2 implies

$$\alpha_1 - \alpha_2 = cv_0$$

for some $c \in \mathbb{R}$. Because both parameter vectors satisfy the same normalization, Assumption 1 implies $c = 0$. Therefore $\alpha_1 = \alpha_2$. $\square$

Proof of Proposition 2. Let A denote the observed bipartite adjacency matrix and let

$$\mathcal{A} = \mathbb{E}[A \mid Z_0, \theta]$$

denote its population counterpart. Under the degree-corrected stochastic block model,

$$\mathcal{A} = \Theta_s Z_{s0} P Z_{t0}' \Theta_t,$$

where Θ_s and Θ_t collect the degree parameters on the two sides of the network.

By Assumption 6, the group proportions are bounded away from zero and the rows and columns of P corresponding to different latent groups are separated. Consequently, the population left and right singular-vector embeddings of $\mathcal{A}$ contain K and L distinct population centers, respectively, one for each latent group.

The minimum-degree condition implies concentration of the sample adjacency matrix around its population counterpart. In particular,

$$\|A - \mathcal{A}\|_{\text{op}} = o_p\left(\sigma_{\min}^+(\mathcal{A})\right),$$

where $\sigma_{\min}^+(\mathcal{A})$ denotes the smallest nonzero singular value relevant for recovering the latent subspaces.

Standard singular-subspace perturbation results therefore imply that, after an appropriate orthogonal transformation, the estimated left and right singular-vector matrices converge to their population counterparts:

$$\|\hat{U} - U_0 O_s\|_F = o_p(\sqrt{N}), \quad \|\hat{V} - V_0 O_t\|_F = o_p(\sqrt{M}).$$

Because the population group centers are separated by strictly positive constants, applying K -means to $\hat{U}$ and L -means to $\hat{V}$ yields vanishing misclassification proportions. Hence there exist permutation matrices Π_s and Π_t such that

$$\frac{1}{N} \|\hat{Z}_s - Z_{s0} \Pi_s\|_0 \xrightarrow{p} 0$$

and

$$\frac{1}{M} \|\hat{Z}_t - Z_{t0} \Pi_t\|_0 \xrightarrow{p} 0.$$

This proves the result. $\square$

Proof of Proposition 3. Define the normalized feasible and infeasible marginal likelihood criteria by

$$Q_n(\psi, \widehat{Z}) = \frac{1}{a_n} \ell_n(\psi; \widehat{Z}), \quad (17)$$

$$Q_n(\psi, Z_0) = \frac{1}{a_n} \ell_n(\psi; Z_0). \quad (18)$$

Recall that the feasible empirical Bayes hyperparameter estimator is

$$\widehat{\psi} = \arg \max_{\psi \in \Psi} Q_n(\psi, \widehat{Z}). \quad (19)$$

We prove that

$$\widehat{\psi} \xrightarrow{P} \psi_0.$$

The proof consists of three steps.

Step 1: Uniform convergence of the feasible criterion.

By the triangle inequality,

$$\begin{aligned} \sup_{\psi \in \Psi} |Q_n(\psi, \widehat{Z}) - Q(\psi)| &\leq \sup_{\psi \in \Psi} |Q_n(\psi, \widehat{Z}) - Q_n(\psi, Z_0)| \\ &\quad + \sup_{\psi \in \Psi} |Q_n(\psi, Z_0) - Q(\psi)|. \end{aligned} \quad (20)$$

By condition (13),

$$\sup_{\psi \in \Psi} |Q_n(\psi, \widehat{Z}) - Q_n(\psi, Z_0)| = o_p(1). \quad (21)$$

Moreover, Assumption 9 implies

$$\sup_{\psi \in \Psi} |Q_n(\psi, Z_0) - Q(\psi)| = o_p(1). \quad (22)$$

Combining (20)–(22) therefore gives

$$\sup_{\psi \in \Psi} |Q_n(\psi, \widehat{Z}) - Q(\psi)| = o_p(1). \quad (23)$$

Hence the feasible marginal likelihood criterion converges uniformly in probability to the same population criterion $Q(\psi)$ as the infeasible criterion based on the true group membership

Z_0 .

Under exact group recovery, condition (21) follows immediately. Indeed, after an appropriate permutation of group labels,

$$\Pr(\widehat{Z} = Z_0) \rightarrow 1.$$

On the event $\{\widehat{Z} = Z_0\}$,

$$Q_n(\psi, \widehat{Z}) = Q_n(\psi, Z_0)$$

for every $\psi \in \Psi$. Therefore,

$$\sup_{\psi \in \Psi} |Q_n(\psi, \widehat{Z}) - Q_n(\psi, Z_0)| = 0$$

with probability approaching one, which implies (21).

Step 2: Separation of the population objective.

Fix an arbitrary $\epsilon > 0$. Define

$$\Psi_\epsilon = \{\psi \in \Psi : \|\psi - \psi_0\| \geq \epsilon\}. \quad (24)$$

Because Ψ is compact, Ψ_ϵ is also compact. By Assumption 7, ψ_0 is the unique identified true parameter, and by Assumption 9, $Q(\psi)$ is uniquely maximized at ψ_0 .

Assume in addition that $Q(\psi)$ is continuous on Ψ . Since Ψ_ϵ is compact, the maximum of Q over Ψ_ϵ exists. Uniqueness of ψ_0 implies

$$\sup_{\psi \in \Psi_\epsilon} Q(\psi) < Q(\psi_0). \quad (25)$$

Define

$$\Delta_\epsilon = Q(\psi_0) - \sup_{\psi \in \Psi_\epsilon} Q(\psi). \quad (26)$$

Then

$$\Delta_\epsilon > 0. \quad (27)$$

This establishes a strictly positive population criterion gap between the true parameter ψ_0 and every parameter lying at least ϵ away from ψ_0 .

Step 3: Consistency of the argmax.

Let

$$\mathcal{A}_n(\epsilon) = \left\{ \sup_{\psi \in \Psi} |Q_n(\psi, \widehat{Z}) - Q(\psi)| < \frac{\Delta_\epsilon}{3} \right\}. \quad (28)$$

By (23),

$$\Pr(\mathcal{A}_n(\epsilon)) \rightarrow 1. \quad (29)$$

On the event $\mathcal{A}_n(\epsilon)$, evaluation of the feasible criterion at the true parameter gives

$$Q_n(\psi_0, \widehat{Z}) > Q(\psi_0) - \frac{\Delta_\epsilon}{3}. \quad (30)$$

For any $\psi \in \Psi_\epsilon$, again on $\mathcal{A}_n(\epsilon)$,

$$\begin{aligned} Q_n(\psi, \widehat{Z}) &< Q(\psi) + \frac{\Delta_\epsilon}{3} \\ &\leq \sup_{\tilde{\psi} \in \Psi_\epsilon} Q(\tilde{\psi}) + \frac{\Delta_\epsilon}{3}. \end{aligned} \quad (31)$$

Using the definition of Δ_ϵ in (26),

$$\begin{aligned} \sup_{\psi \in \Psi_\epsilon} Q_n(\psi, \widehat{Z}) &< Q(\psi_0) - \Delta_\epsilon + \frac{\Delta_\epsilon}{3} \\ &= Q(\psi_0) - \frac{2\Delta_\epsilon}{3}. \end{aligned} \quad (32)$$

On the other hand, (30) gives

$$Q_n(\psi_0, \widehat{Z}) > Q(\psi_0) - \frac{\Delta_\epsilon}{3}. \quad (33)$$

Since

$$Q(\psi_0) - \frac{\Delta_\epsilon}{3} > Q(\psi_0) - \frac{2\Delta_\epsilon}{3},$$

we obtain

$$Q_n(\psi_0, \widehat{Z}) > \sup_{\psi \in \Psi_\epsilon} Q_n(\psi, \widehat{Z}) \quad (34)$$

on $\mathcal{A}_n(\epsilon)$.

Therefore, on $\mathcal{A}_n(\epsilon)$, a maximizer of the feasible criterion cannot belong to Ψ_ϵ . By (19),

$$\widehat{\psi} \notin \Psi_\epsilon. \quad (35)$$

Equivalently,

$$\|\widehat{\psi} - \psi_0\| < \epsilon. \quad (36)$$

It follows that

$$\begin{aligned} \Pr \left(\|\widehat{\psi} - \psi_0\| \geq \epsilon \right) &\leq \Pr \left(\mathcal{A}_n(\epsilon)^c \right) \\ &\longrightarrow 0. \end{aligned} \quad (37)$$

Since $\epsilon > 0$ was arbitrary,

$$\boxed{\widehat{\psi} \xrightarrow{p} \psi_0.} \quad (38)$$

Finally, recall that

$$\psi = \left(\mu', (\tau^2)', \sigma_\varepsilon^2 \right)'.$$

Convergence of the finite-dimensional vector $\widehat{\psi}$ therefore implies componentwise convergence.

Hence, after aligning the estimated group labels with their population labels,

$$\widehat{\mu}_k^s \xrightarrow{p} \mu_{k,0}^s, \quad \widehat{\tau}_{k,s}^2 \xrightarrow{p} \tau_{k,s,0}^2, \quad k = 1, \dots, K, \quad (39)$$

$$\widehat{\mu}_\ell^t \xrightarrow{p} \mu_{\ell,0}^t, \quad \widehat{\tau}_{\ell,t}^2 \xrightarrow{p} \tau_{\ell,t,0}^2, \quad \ell = 1, \dots, L, \quad (40)$$

and

$$\widehat{\sigma}_\varepsilon^2 \xrightarrow{p} \sigma_{\varepsilon,0}^2. \quad (41)$$

This establishes the stated consistency result. $\square$

Proof of Proposition 4. Throughout the proof, group labels are aligned with their population counterparts. This is without loss of generality because latent-group labels are identified only up to permutation.

Let

$$p_n = N + M,$$

and define

$$m_0 = Z_0 \mu_0, \quad \widehat{m} = \widehat{Z} \widehat{\mu},$$

as well as

$$D_0 = D(\tau_0^2), \quad \widehat{D} = D(\widehat{\tau}^2).$$

The oracle empirical Bayes estimator is

$$\alpha^{EB,O} = m_0 + D_0 B' \left(B D_0 B' + \sigma_{\varepsilon,0}^2 I_n \right)^{-1} \left(\tilde{Y} - B m_0 \right), \quad (42)$$

whereas the feasible estimator is

$$\hat{\alpha}^{EB} = \hat{m} + \hat{D} B' \left(B \hat{D} B' + \hat{\sigma}_{\varepsilon}^2 I_n \right)^{-1} \left(\tilde{Y} - B \hat{m} \right). \quad (43)$$

We establish the result in five steps.

Step 1: Convergence of the estimated prior mean vector.

By exact group recovery,

$$\Pr(\hat{Z} = Z_0) \rightarrow 1$$

after aligning the group labels. Hence, with probability approaching one,

$$\hat{m} - m_0 = Z_0(\hat{\mu} - \mu_0).$$

Since every row of Z_0 contains exactly one nonzero element,

$$\begin{aligned} \frac{1}{p_n} \|\hat{m} - m_0\|^2 &= \frac{1}{p_n} \sum_{r=1}^{K+L} n_r (\hat{\mu}_r - \mu_{r,0})^2 \\ &\leq \max_{1 \leq r \leq K+L} |\hat{\mu}_r - \mu_{r,0}|^2, \end{aligned} \quad (44)$$

where n_r denotes the number of nodes belonging to group r .

Because $K + L$ is fixed and Proposition 3 implies

$$\hat{\mu}_r \xrightarrow{p} \mu_{r,0}$$

for every r ,

$$\max_{1 \leq r \leq K+L} |\hat{\mu}_r - \mu_{r,0}| = o_p(1). \quad (45)$$

Therefore,

$$\frac{1}{p_n} \|\hat{m} - m_0\|^2 = o_p(1). \quad (46)$$

Step 2: Convergence of the prior precision matrices.

Define

$$\Lambda_0 = \sigma_{\varepsilon,0}^2 D_0^{-1} \quad (47)$$

and

$$\hat{\Lambda} = \hat{\sigma}_\varepsilon^2 \hat{D}^{-1}. \quad (48)$$

Under exact group recovery, both D_0 and $\hat{D}$ are diagonal matrices whose diagonal elements correspond to the same latent groups. Consequently,

$$\|\hat{D} - D_0\|_{\text{op}} = \max_{1 \leq r \leq K+L} \left| \hat{\tau}_r^2 - \tau_{r,0}^2 \right|. \quad (49)$$

By Proposition 3,

$$\hat{\tau}_r^2 \xrightarrow{p} \tau_{r,0}^2$$

for every r . Since the number of groups is fixed,

$$\|\hat{D} - D_0\|_{\text{op}} = o_p(1). \quad (50)$$

The prior variances are bounded away from zero by assumption. Therefore matrix inversion is continuous in a neighborhood of D_0 , and

$$\|\hat{D}^{-1} - D_0^{-1}\|_{\text{op}} = o_p(1). \quad (51)$$

Moreover,

$$\hat{\sigma}_\varepsilon^2 \xrightarrow{p} \sigma_{\varepsilon,0}^2.$$

Hence

$$\begin{aligned} \|\hat{\Lambda} - \Lambda_0\|_{\text{op}} &= \left\| \hat{\sigma}_\varepsilon^2 \hat{D}^{-1} - \sigma_{\varepsilon,0}^2 D_0^{-1} \right\|_{\text{op}} \\ &\leq \left| \hat{\sigma}_\varepsilon^2 - \sigma_{\varepsilon,0}^2 \right| \|\hat{D}^{-1}\|_{\text{op}} \\ &\quad + \sigma_{\varepsilon,0}^2 \|\hat{D}^{-1} - D_0^{-1}\|_{\text{op}} \\ &= o_p(1). \end{aligned} \quad (52)$$

Step 3: Stability of the empirical Bayes updating matrix.

Using the standard matrix identity

$$DB' \left(BDB' + \sigma_\varepsilon^2 I_n \right)^{-1} = \left(B'B + \sigma_\varepsilon^2 D^{-1} \right)^{-1} B', \quad (53)$$

define

$$A_0 = (B'B + \Lambda_0)^{-1} B', \quad (54)$$

$$\hat{A} = (B'B + \hat{\Lambda})^{-1} B'. \quad (55)$$

Let

$$H_0 = B'B + \Lambda_0, \quad \hat{H} = B'B + \hat{\Lambda}.$$

Since $B'B$ is positive semidefinite,

$$\lambda_{\min}(H_0) \geq \lambda_{\min}(\Lambda_0).$$

By the uniform lower and upper bounds imposed on $\tau_{r,0}^2$ and $\sigma_{\varepsilon,0}^2$, there exists a constant $c_\Lambda > 0$ such that

$$\lambda_{\min}(\Lambda_0) \geq c_\Lambda. \quad (56)$$

Therefore

$$\|H_0^{-1}\|_{\text{op}} \leq c_\Lambda^{-1}. \quad (57)$$

By (52), with probability approaching one,

$$\lambda_{\min}(\hat{\Lambda}) \geq \frac{c_\Lambda}{2},$$

so

$$\|\hat{H}^{-1}\|_{\text{op}} = O_p(1). \quad (58)$$

Using the resolvent identity,

$$\hat{H}^{-1} - H_0^{-1} = \hat{H}^{-1} (\Lambda_0 - \hat{\Lambda}) H_0^{-1}. \quad (59)$$

It follows that

$$\begin{aligned}
\hat{A} - A_0 &= (\hat{H}^{-1} - H_0^{-1}) B' \\
&= \hat{H}^{-1} (\Lambda_0 - \hat{\Lambda}) H_0^{-1} B' \\
&= \hat{H}^{-1} (\Lambda_0 - \hat{\Lambda}) A_0.
\end{aligned} \tag{60}$$

We next show that $\|A_0\|_{\text{op}}$ is bounded. Observe that

$$H_0 = B'B + \Lambda_0 \succeq B'B.$$

Therefore

$$H_0^{-1/2} B' B H_0^{-1/2} \preceq I_{p_n}. \tag{61}$$

Consequently,

$$\begin{aligned}
\|A_0\|_{\text{op}} &= \|H_0^{-1} B'\|_{\text{op}} \\
&\leq \|H_0^{-1/2}\|_{\text{op}} \|H_0^{-1/2} B'\|_{\text{op}} \\
&\leq c_{\Lambda}^{-1/2}.
\end{aligned} \tag{62}$$

Combining (52), (58), (60), and (62), we obtain

$$\|\hat{A} - A_0\|_{\text{op}} = o_p(1). \tag{63}$$

We will also require convergence of the corresponding shrinkage operators. Since

$$A_0 B = H_0^{-1} B' B = I - H_0^{-1} \Lambda_0,$$

and similarly

$$\hat{A} B = I - \hat{H}^{-1} \hat{\Lambda},$$

we have

$$\begin{aligned}
\hat{A} B - A_0 B &= H_0^{-1} \Lambda_0 - \hat{H}^{-1} \hat{\Lambda} \\
&= (H_0^{-1} - \hat{H}^{-1}) \Lambda_0 + \hat{H}^{-1} (\Lambda_0 - \hat{\Lambda}).
\end{aligned} \tag{64}$$

Using (59) and the boundedness established above,

$$\|\hat{A}B - A_0B\|_{\text{op}} = o_p(1). \quad (65)$$

Moreover,

$$I - \hat{A}B = \hat{H}^{-1}\hat{\Lambda}, \quad (66)$$

so

$$\|I - \hat{A}B\|_{\text{op}} = O_p(1). \quad (67)$$

Step 4: Decomposition of the feasible-oracle difference.

Using the compact representations

$$\hat{\alpha}^{EB} = \hat{m} + \hat{A}(\tilde{Y} - B\hat{m}), \quad (68)$$

$$\alpha^{EB,O} = m_0 + A_0(\tilde{Y} - Bm_0), \quad (69)$$

we obtain

$$\begin{aligned} \hat{\alpha}^{EB} - \alpha^{EB,O} &= \hat{m} - m_0 - \hat{A}B(\hat{m} - m_0) \\ &\quad + (\hat{A} - A_0)(\tilde{Y} - Bm_0) \\ &= (I - \hat{A}B)(\hat{m} - m_0) \\ &\quad + (\hat{A} - A_0)(\tilde{Y} - Bm_0). \end{aligned} \quad (70)$$

Since

$$\tilde{Y} = B\alpha + \varepsilon,$$

we have

$$\tilde{Y} - Bm_0 = B(\alpha - m_0) + \varepsilon.$$

Hence

$$\begin{aligned} \hat{\alpha}^{EB} - \alpha^{EB,O} &= (I - \hat{A}B)(\hat{m} - m_0) \\ &\quad + (\hat{A} - A_0)B(\alpha - m_0) \\ &\quad + (\hat{A} - A_0)\varepsilon. \end{aligned} \quad (71)$$

We treat the three terms separately.

Step 5: Each component is asymptotically negligible.

For the first term, using (67) and (46),

$$\begin{aligned}
\frac{1}{p_n} \left\| (I - \hat{A}B)(\hat{m} - m_0) \right\|^2 &\leq \|I - \hat{A}B\|_{\text{op}}^2 \frac{\|\hat{m} - m_0\|^2}{p_n} \\
&= O_p(1) o_p(1) \\
&= o_p(1).
\end{aligned} \tag{72}$$

For the second term, from (65),

$$\begin{aligned}
\frac{1}{p_n} \left\| (\hat{A} - A_0)B(\alpha - m_0) \right\|^2 &= \frac{1}{p_n} \left\| (\hat{A}B - A_0B)(\alpha - m_0) \right\|^2 \\
&\leq \|\hat{A}B - A_0B\|_{\text{op}}^2 \frac{\|\alpha - m_0\|^2}{p_n}.
\end{aligned} \tag{73}$$

By assumption,

$$\frac{1}{p_n} \|\alpha - m_0\|^2 = O_p(1),$$

while

$$\|\hat{A}B - A_0B\|_{\text{op}} = o_p(1).$$

Therefore

$$\frac{1}{p_n} \left\| (\hat{A} - A_0)B(\alpha - m_0) \right\|^2 = o_p(1). \tag{74}$$

For the third term, using (60),

$$(\hat{A} - A_0)\varepsilon = \hat{H}^{-1}(\Lambda_0 - \hat{\Lambda})A_0\varepsilon. \tag{75}$$

Hence

$$\frac{1}{p_n} \|(\hat{A} - A_0)\varepsilon\|^2 \leq \|\hat{H}^{-1}\|_{\text{op}}^2 \|\hat{\Lambda} - \Lambda_0\|_{\text{op}}^2 \frac{\|A_0\varepsilon\|^2}{p_n}. \tag{76}$$

It remains to show that

$$\frac{1}{p_n} \|A_0\varepsilon\|^2 = O_p(1).$$

Conditional on the network B ,

$$\begin{aligned}\mathbb{E} \left[\frac{1}{p_n} \|A_0 \varepsilon\|^2 \mid B \right] &= \frac{\sigma_{\varepsilon,0}^2}{p_n} \text{tr} (A_0 A_0') \\ &\leq \sigma_{\varepsilon,0}^2 \|A_0\|_{\text{op}}^2 \\ &\leq \sigma_{\varepsilon,0}^2 c_{\Lambda}^{-1}.\end{aligned}\tag{77}$$

The final expression is uniformly bounded. By Markov's inequality,

$$\frac{1}{p_n} \|A_0 \varepsilon\|^2 = O_p(1).\tag{78}$$

Combining (58), (52), and (78) in (76) yields

$$\frac{1}{p_n} \|(\hat{A} - A_0) \varepsilon\|^2 = o_p(1).\tag{79}$$

Finally, using

$$\|a + b + c\|^2 \leq 3\|a\|^2 + 3\|b\|^2 + 3\|c\|^2$$

together with (72), (74), and (79), we obtain

$$\frac{1}{p_n} \left\| \hat{\alpha}^{EB} - \alpha^{EB,O} \right\|^2 = o_p(1).\tag{80}$$

Since $p_n = N + M$,

$$\boxed{\frac{1}{N + M} \left\| \hat{\alpha}^{EB} - \alpha^{EB,O} \right\|^2 \xrightarrow{p} 0.}\tag{81}$$

This proves the proposition. $\square$

Proof of Proposition 5. All calculations below are carried out on the normalized identified subspace, so that the zero eigenvalue associated with the location indeterminacy is excluded.

Under the scalar-prior specification

$$D = \tau^2 I,$$

the empirical Bayes posterior mean can be equivalently written as

$$\hat{\alpha}^{EB} = m + (Q_B + \eta I)^{-1} B' (\tilde{Y} - Bm),\tag{82}$$

where

$$Q_B = B'B, \quad \eta = \frac{\sigma_\varepsilon^2}{\tau^2}.$$

Using

$$\tilde{Y} = B\alpha + \varepsilon,$$

we obtain

$$\begin{aligned} \hat{\alpha}^{EB} - \alpha &= m - \alpha + (Q_B + \eta I)^{-1} B' [B(\alpha - m) + \varepsilon] \\ &= m - \alpha + (Q_B + \eta I)^{-1} Q_B (\alpha - m) + (Q_B + \eta I)^{-1} B' \varepsilon. \end{aligned} \quad (83)$$

Since

$$(Q_B + \eta I)^{-1} Q_B = I - \eta(Q_B + \eta I)^{-1},$$

equation (83) reduces to

$$\hat{\alpha}^{EB} - \alpha = -\eta(Q_B + \eta I)^{-1}(\alpha - m) + (Q_B + \eta I)^{-1} B' \varepsilon. \quad (84)$$

The first term in (84) is the shrinkage-bias component and the second term is the sampling-error component.

Let the eigendecomposition of Q_B be

$$Q_B = VQV', \quad (85)$$

where

$$Q = \text{diag}(q_1, \dots, q_{N+M})$$

and $V'V = I$. Define

$$b = V'(\alpha - m).$$

Conditional on the realized heterogeneous effects α , the error ε has mean zero. Hence

$$\mathbb{E} [\hat{\alpha}^{EB} - \alpha \mid \alpha] = -\eta(Q_B + \eta I)^{-1}(\alpha - m). \quad (86)$$

The squared conditional bias is therefore

$$\begin{aligned}\left\|\mathbb{E}\left[\hat{\alpha}^{EB} - \alpha \mid \alpha\right]\right\|^2 &= \eta^2(\alpha - m)'(Q_B + \eta I)^{-2}(\alpha - m) \\ &= \eta^2 b'(Q + \eta I)^{-2}b.\end{aligned}\tag{87}$$

Restricting attention to the identified subspace gives

$$\text{Bias}_{EB}^2(\alpha) = \sum_{r=2}^{N+M} \frac{\eta^2 b_r^2}{(q_r + \eta)^2}.\tag{88}$$

Next consider the variance term. Conditional on α ,

$$\text{Var}\left(\hat{\alpha}^{EB} \mid \alpha\right) = \sigma_\varepsilon^2 (Q_B + \eta I)^{-1} B' B (Q_B + \eta I)^{-1}.\tag{89}$$

Since $B' B = Q_B$,

$$\text{Var}\left(\hat{\alpha}^{EB} \mid \alpha\right) = \sigma_\varepsilon^2 (Q_B + \eta I)^{-1} Q_B (Q_B + \eta I)^{-1}.\tag{90}$$

Taking the trace and using the spectral decomposition yields

$$\begin{aligned}\text{tr}\left[\text{Var}\left(\hat{\alpha}^{EB} \mid \alpha\right)\right] &= \sigma_\varepsilon^2 \text{tr}\left[(Q + \eta I)^{-1} Q (Q + \eta I)^{-1}\right] \\ &= \sigma_\varepsilon^2 \sum_{r=2}^{N+M} \frac{q_r}{(q_r + \eta)^2}.\end{aligned}\tag{91}$$

Combining (88) and (91), the conditional mean squared error of the empirical Bayes estimator is

$$\begin{aligned}\text{MSE}_{EB}(\alpha) &= \sum_{r=2}^{N+M} \frac{\eta^2 b_r^2}{(q_r + \eta)^2} \\ &\quad + \sigma_\varepsilon^2 \sum_{r=2}^{N+M} \frac{q_r}{(q_r + \eta)^2}.\end{aligned}\tag{92}$$

We next derive the corresponding MSE for the normalized least-squares estimator. On the identified subspace,

$$\hat{\alpha}^{OLS} = Q_B^+ B' \tilde{Y},\tag{93}$$

where Q_B^+ denotes the Moore–Penrose inverse of Q_B , restricted to the normalized parameter space. Because

$$\tilde{Y} = B\alpha + \varepsilon,$$

and

$$Q_B^+ Q_B \alpha = \alpha$$

on this subspace,

$$\hat{\alpha}^{OLS} - \alpha = Q_B^+ B' \varepsilon. \quad (94)$$

Therefore OLS is conditionally unbiased:

$$\mathbb{E} [\hat{\alpha}^{OLS} - \alpha \mid \alpha] = 0. \quad (95)$$

Its conditional covariance matrix is

$$\begin{aligned} \text{Var}(\hat{\alpha}^{OLS} \mid \alpha) &= \sigma_\varepsilon^2 Q_B^+ B' B Q_B^+ \\ &= \sigma_\varepsilon^2 Q_B^+. \end{aligned} \quad (96)$$

Consequently,

$$\text{MSE}_{OLS} = \sigma_\varepsilon^2 \sum_{r=2}^{N+M} \frac{1}{q_r}. \quad (97)$$

We can now compare the two estimators. From (92) and (97),

$$\begin{aligned} &\text{MSE}_{OLS} - \text{MSE}_{EB}(\alpha) \\ &= \sigma_\varepsilon^2 \sum_{r=2}^{N+M} \frac{1}{q_r} - \sigma_\varepsilon^2 \sum_{r=2}^{N+M} \frac{q_r}{(q_r + \eta)^2} - \sum_{r=2}^{N+M} \frac{\eta^2 b_r^2}{(q_r + \eta)^2} \\ &= \sigma_\varepsilon^2 \sum_{r=2}^{N+M} \left[\frac{1}{q_r} - \frac{q_r}{(q_r + \eta)^2} \right] - \sum_{r=2}^{N+M} \frac{\eta^2 b_r^2}{(q_r + \eta)^2}. \end{aligned} \quad (98)$$

Hence

$$\text{MSE}_{EB}(\alpha) \leq \text{MSE}_{OLS}$$

if and only if

$$\sum_{r=2}^{N+M} \frac{\eta^2 b_r^2}{(q_r + \eta)^2} \leq \sigma_\varepsilon^2 \sum_{r=2}^{N+M} \left[\frac{1}{q_r} - \frac{q_r}{(q_r + \eta)^2} \right]. \quad (99)$$

This is precisely condition (17), proving the stated necessary and sufficient condition. $\square$

Appendix B. Supplementary Material

SM.1.1 Determine the number of Groups

We first use the observed user–product interaction network to recover latent groups of users and products. Importantly, the group-detection procedure is based only on whether a user has reviewed a product and does not use the numerical rating outcome. This separation ensures that the dependent variable in the subsequent empirical Bayes analysis does not mechanically determine the latent group memberships. Let

$$B \in \{0, 1\}^{N \times M}$$

denote the rectangular biadjacency matrix of the bipartite network, where N is the number of users and M is the number of products. Its (i, j) -th element is defined as

$$B_{ij} = \begin{cases} 1, & \text{if user } i \text{ rated product } j, \\ 0, & \text{otherwise.} \end{cases}$$

In the empirical application, the network consists of $N = 50,985$ users, $M = 25,848$ products, and $E = 412,947$ observed user–product links. The degree of user i and the degree of product j are respectively defined as

$$d_i = \sum_{j=1}^M B_{ij}, \quad q_j = \sum_{i=1}^N B_{ij}.$$

These degrees vary substantially across nodes, reflecting pronounced heterogeneity in user activity and product popularity.

To prevent such degree heterogeneity from being mistaken for genuine community structure, we employ a degree-corrected bipartite stochastic block model. The specification is motivated by the degree-corrected stochastic block model of ?, adapted here to a network with two distinct sets of nodes. Let

$$g_i \in \{1, \dots, K\}$$

denote the latent group membership of user i , and let

$$h_j \in \{1, \dots, L\}$$

denote the latent group membership of product j . In contrast to a one-mode block model, the numbers of groups on the two sides of the network are not required to be equal. Under the Poisson formulation of the degree-corrected model, the conditional edge intensity is

$$\mathbb{E}[B_{ij} \mid g_i, h_j] = \theta_i \phi_j \omega_{g_i h_j},$$

where θ_i captures the activity propensity of user i , ϕ_j captures the popularity propensity of product j , and ω_{kl} measures the underlying matching affinity between user group k and product group l . Thus, the expected interaction intensity can be interpreted as the product of user activity, product popularity, and group-level matching affinity.

The multiplicative representation requires a normalization because the degree parameters and block-affinity parameters are otherwise not separately identified. We impose the within-group normalizations

$$\sum_{i: g_i=k} \theta_i = 1, \quad k = 1, \dots, K,$$

and

$$\sum_{j: h_j=l} \phi_j = 1, \quad l = 1, \dots, L.$$

Define the total degree associated with user group k and product group l as

$$\kappa_k = \sum_{i: g_i=k} d_i, \quad \eta_l = \sum_{j: h_j=l} q_j,$$

and define the observed number of links between the corresponding pair of groups as

$$e_{kl} = \sum_{i: g_i=k} \sum_{j: h_j=l} B_{ij}.$$

Conditional on a given partition, the maximum-likelihood estimates of the node-specific degree parameters take the natural form

$$\hat{\theta}_i = \frac{d_i}{\kappa_{g_i}}, \quad \hat{\phi}_j = \frac{q_j}{\eta_{h_j}},$$

while, under the normalization above, the block-intensity estimator is proportional to the corresponding block-edge count e_{kl} . Consequently, heterogeneity in the overall activity of individual users and the popularity of individual products is absorbed by the node-specific parameters

rather than by the latent group assignments.

Profiling out the degree parameters yields a convenient objective for evaluating a candidate partition. Let

$$E = \sum_{i=1}^N \sum_{j=1}^M B_{ij}$$

denote the total number of observed edges. Conditional on the degree sequence, the expected number of edges between user group k and product group l under random bipartite matching is

$$e_{kl}^{(0)} = \frac{\kappa_k \eta_l}{E}.$$

We therefore evaluate a candidate partition using the degree-corrected profile objective

$$Q_{\text{DC}}(K, L) = \sum_{k=1}^K \sum_{l=1}^L e_{kl} \log \left(\frac{e_{kl}}{e_{kl}^{(0)}} \right) = \sum_{k=1}^K \sum_{l=1}^L e_{kl} \log \left(\frac{e_{kl} E}{\kappa_k \eta_l} \right),$$

where a term with $e_{kl} = 0$ is defined to be zero. The factor E inside the logarithm contributes only the constant $E \log E$ across candidate partitions and therefore does not affect the ranking of alternative group assignments. Economically, Q_{DC} measures the extent to which observed user-group–product-group interactions differ from what would be expected solely from heterogeneous user activity and product popularity.

Direct maximization of the block-model likelihood over all possible user and product assignments is computationally infeasible for a network of this size. We therefore estimate the latent memberships using degree-normalized spectral co-clustering, following the general bipartite spectral-clustering logic of ? and the normalized asymmetric clustering framework of ?. Let

$$D_u = \text{diag}(d_1, \dots, d_N), \quad D_p = \text{diag}(q_1, \dots, q_M),$$

and construct the normalized biadjacency matrix

$$\tilde{B} = D_u^{-1/2} B D_p^{-1/2}.$$

We then compute its truncated singular value decomposition,

$$\tilde{B} = U \Sigma V^\top.$$

The left singular vectors contain the spectral information associated with users, whereas the right singular vectors contain the corresponding information for products. For the normalized network, the leading singular direction primarily represents the global degree component and is therefore excluded from the clustering step.

For each candidate pair (K, L) , we define

$$r = \min\{K, L\},$$

and construct the nontrivial user and product embeddings as

$$U^* = [u_2, \dots, u_r], \quad V^* = [v_2, \dots, v_r].$$

The rows of U^* and V^* are normalized to unit Euclidean length before clustering. We then cluster the rows of U^* into K user groups and independently cluster the rows of V^* into L product groups. In the implementation, K -means-type clustering is performed separately on the two embeddings, with MiniBatch K -means used for computational scalability. Because the K -means objective can be sensitive to initialization, several independent initializations are considered for each candidate pair (K, L) . For a fixed (K, L) , we retain the user and product assignments that produce the largest value of $Q_{\text{DC}}(K, L)$.

Although the degree-corrected profile objective increases the fit of the model, it cannot by itself determine the appropriate numbers of groups, because larger values of K and L generally provide greater flexibility. We therefore introduce a complexity penalty motivated by the corrected Bayesian information criterion of ?. Their one-mode CBIC explicitly penalizes the latent community assignments in addition to the usual model-parameter complexity. In a one-mode network with n nodes and k groups, there are approximately k^n possible membership configurations, whose logarithm is $n \log k$. In the present bipartite setting, the user side has approximately K^N possible assignments and the product side has approximately L^M possible assignments. Hence, their combined log assignment complexity is

$$N \log K + M \log L.$$

Because our network is rectangular, the block-affinity matrix contains KL parameters rather than the $k(k+1)/2$ parameters arising from a symmetric one-mode block matrix. Motivated by

this distinction, we use the following degree-corrected bipartite adaptation of the CBIC:

$$\text{CBIC}_{\text{DC}}(K, L) = -2Q_{\text{DC}}(K, L) + 2N \log K + 2M \log L + KL \log(NM).$$

The first term rewards goodness of fit, the second and third terms penalize the uncertainty associated with user and product membership assignments, respectively, and the final term penalizes the KL block-affinity parameters. We emphasize that this exact rectangular expression is a bipartite adaptation of the CBIC principle rather than a formula stated verbatim in ?. The numbers of user and product groups are therefore selected jointly according to

$$(\widehat{K}, \widehat{L}) = \arg \min_{K, L} \text{CBIC}_{\text{DC}}(K, L).$$

In the empirical implementation, we first examine a broad set of candidate values for K and L , and subsequently conduct a finer search around the region attaining the smallest criterion values. Multiple clustering initializations are evaluated for each candidate specification, and the partition attaining the highest degree-corrected profile objective is used to calculate the corresponding CBIC. This procedure selects

$$\widehat{K} = 10, \quad \widehat{L} = 11,$$

indicating that the latent structure of the user side and product side cannot be adequately represented by imposing an equal number of groups. Because cluster labels are arbitrary up to permutation, the estimated block structure is not required to be diagonal: for example, user group k may have its strongest interaction affinity with product group $l \neq k$. Accordingly, the economically relevant objects are the estimated group-to-group affinities rather than the numerical labels assigned to the groups.

Finally, the estimated memberships are used as inputs to the empirical Bayes estimation of heterogeneous user and product effects. Let

$$\widehat{Z}_u \in \{0, 1\}^{N \times \widehat{K}}$$

and

$$\widehat{Z}_p \in \{0, 1\}^{M \times \widehat{L}}$$

denote the estimated one-hot membership matrices for users and products, respectively. We combine them into

$$\hat{Z} = \begin{pmatrix} \hat{Z}_u & 0 \\ 0 & \hat{Z}_p \end{pmatrix} \in \{0, 1\}^{(N+M) \times (\hat{K} + \hat{L})}.$$

Since $\hat{K} = 10$ and $\hat{L} = 11$, the subsequent empirical Bayes analysis contains 21 group-specific distributions of heterogeneous effects. The estimated network memberships therefore determine which users and products are pooled together when learning the group-specific prior distributions, while the observed rating equation is subsequently used to estimate the magnitude of individual user and product effects. “

SM.2.2 Classification in Empirical Study